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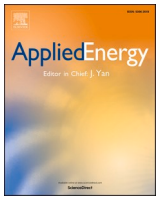
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Advancing an already high-performance smart building with model predictive control: Multi-layer optimization under forecast uncertainty in a real building case

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HIGHLIGHTS

- How can MPC with a novel cost function improve an already efficient thermal system?
- Real-time MPC via TRNSYS–C++ co-simulation using PAGMO and Differential Evolution.
- 13 % operational cost savings and 85 % TES degradation reduction under uncertainty
- MPC impact relies on system flexibility and is strongest in transitional seasons.
- Long-term NPV nearly doubled; payback shortened by 4+ years vs. rule-based control.

ARTICLE INFO

Keywords:
Model predictive control (MPC)
Forecast uncertainty
Ground source heat pump
Cost penalty optimization
Smart HVAC
Boreholes
Building decarbonization

ABSTRACT

Thermal energy systems in buildings play a central role in global decarbonization efforts, accounting for a significant share of energy use and carbon emissions. This study addresses a key research question: how can advanced control strategies further enhance the performance of already energy-efficient, low-exergy thermal systems in low-energy buildings? To address this, a model predictive control (MPC) framework is designed to optimize the operation of an advanced thermal system based on modern concepts of low-temperature heating and high-temperature cooling, including ground-source heat pumps, borehole thermal storage, and modern air handling units. This approach employs a multi-layered MPC cost function, considering both immediate operational costs (electricity and heating) as well as system impact penalties, such as CO_2 emissions, thermal energy storage preservation, comfort violations, and peak load shaving, in response to fluctuating market cost signals, outdoor temperature, and thermal storage limitations. Applied to a validated, ultra-efficient commercial building, the MPC framework achieves a 13 % reduction in annual market-responsive operational costs, a 20 % improvement in long-term savings, and a four-year shorter payback period compared to existing well-established rule-based control. The results further confirm the robustness of predictive control under realistic forecast errors, as demonstrated by Monte Carlo simulations. From an environmental perspective, the CO_2 emission index stays below both Swedish electricity and district heating baselines, demonstrating the environmental benefits of predictive control through strategic sector coupling. Beyond the case study, the proposed method provides a scalable pathway for integrating predictive control into next-generation smart buildings. It highlights the potential of MPC as the final optimization layer in advanced thermal systems, aligning with global objectives for cost-promising and carbon-neutral building operations.

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1. Introduction

The building sector accounts for a significant share of global final energy use, with approximately 50–55 % of that usage devoted to meeting thermal demands for heating and cooling [1]. An increase in this demand is expected over the coming decades as temperature extremes are driven by climate change, urbanization, and lifestyle shifts [2]. Therefore, a corresponding rise in greenhouse gas emissions is anticipated, jeopardizing the achievement of global climate targets and decarbonization objectives. In many regions, the energy mix for space heating and cooling remains dominated by fossil fuels, thereby exacerbating environmental concerns [3]. The sustainability and efficiency of thermal-system design in buildings are thus regarded as both an urgent challenge and a critical opportunity for effective climate action [4].

To address these challenges, growing attention has been directed toward low-temperature heating (LTH) and high-temperature cooling (HTC) systems. In LTH applications, supply temperatures are maintained below 45 °C, whereas chilled-water temperatures of approximately 16–18 °C are employed for HTC systems [5]. Multiple advantages are provided: energy efficiency is improved, compatibility with renewable energy sources is increased, thermal distribution losses in networks are reduced, and a greater potential for integrating thermal energy storage (TES) is offered. Additionally, moderate temperature levels facilitate efficient heat pump operation, enabling direct heat exchange with the ground or ambient environment. Despite the established thermodynamic advantages of these systems, their practical performance is limited by traditional, rule-based control strategies that do not fully utilize their dynamic flexibility. This has established a significant research need for advanced control frameworks, such as MPC, that can optimize system operations in response to varying market signals, changing climatic conditions, and storage constraints.

Several researchers have investigated the energy and comfort benefits of LTH and HTC systems. Hesaraki and Huda [7] concluded that LTH-HTC systems can realize energy savings ranging from 10 % to 30 % compared to traditional air-based systems. Liu et al. [8] concluded that minimizing temperature differences between the thermal supply and comfort, as well as mixing losses, can substantially improve HVAC efficiency. Ryu and Lee [9] examined the feasibility of an LTH system within a South Korean apartment complex featuring decentralized substations. Annual heat loss reduction ranged from 4.8 % to 8.9 % when the supply temperature decreased to 55 °C and 50 °C. Using a ground-source heat pump, Sârbu and Dorca [6] achieved an exceptional coefficient of performance (COP) of 5.45, with the lowest carbon dioxide emissions. González-Cruz and Krüger [10] designed a radiant-capacitive heating and cooling system that utilizes passive energy sources to minimize peak power consumption. Bouacida et al. [11] developed a cost-effective radiant ceiling system for the climate in Algeria. Their design achieved an 89 % reduction in cooling energy use, primarily due to the higher cooling supply temperature and the use of simple materials. Arghand et al. [12] compared an HTC system based on direct cooling with a traditional chiller-based system in an office building in Gothenburg, Sweden. They showed that the proposed HTC system is economically superior due to the reduced operational costs. Despite these promising results, the majority of studies have concentrated on component-level or short-term performance, neglecting system-wide dynamics and the impact of operational control strategies. Limited research has investigated the performance of LTH–HTC systems under variable energy tariffs, as well as the potential for enhancement through real-time optimization frameworks. In addition, limited investigations have focused on the control-oriented challenges or system-level optimization necessary to attain such performance in real operating conditions, an area this study aims to address.

Ground-source energy, particularly when implemented through borehole thermal-energy storage (TES), is regarded as one of the most effective and widely adopted renewable options for supplying LTH and HTC due to a stable, seasonally moderated subsurface temperature [13,14]. In Sweden, where well-insulated buildings are prevalent, borehole TES is routinely incorporated into advanced HVAC configurations. Supportive regulations, mature drilling technologies, and extensive experience with ground-source energy systems have established national leadership in this field, delivering cost-effective, emission-free heating and cooling throughout extended operational lifetimes. Ekmekci et al. [15] concluded that using borehole TES for residential buildings in the city of Uppsala, Sweden, considerably eliminates operational costs and emissions with an acceptable payback time of 14 years. Siiskonen [16] assessed the possibility of storing excess heat from district heating during summer heat in borehole TES to enhance heat pump efficiency for winter operation in an LTH system in Gothenburg. Arghand et al. [17] reported that for an advanced HVAC system based on borehole TES for LTH and HTC supply, if the ground temperature is properly balanced, the addition of a heat pump results in a desirable energy cost reduction and profit in a Swedish commercial building with a total area of approximately 1700 m^2. Ryan et al. [18] compared the techno-economic metrics of a borehole-driven heat pump with a combined heat and power plant in a cold climate. According to their outcomes, a properly sized TES tank reduces peak demand by about 22 % while cutting the costs by up to 5 %. In another study, Behzadi et al. [19] demonstrated that more than 81 % of the annual demands are met independently of the grid through a rule-based control framework in a geothermal-driven LTH-HTC system integrated with a reversible heat pump for a residential district in Stockholm, Sweden. These studies affirm the significant technical potential of borehole TES. However, the majority depend on rule-based or static operational strategies, which restrict adaptability to fluctuating energy prices or seasonal variations in storage. Issues, including prolonged ground-temperature drift and the lack of optimized charging and discharging control, have not been adequately addressed.

The performance potential of LTH-HTC systems integrated with borehole TES can be further improved by implementing advanced control strategies. Conventional feedback-based control methods, including thermostatic regulation and rule-based logic, rely on reactive adjustments and often fail to fully leverage the flexibility that thermal storage and the dynamics of energy pricing could offer. In contrast, feedforward control methods, particularly MPC, offer more advanced and proactive control options [20,21]. MPC predicts future system states using forecast data such as weather, occupancy, and electricity prices, and optimizes control actions across a prediction horizon. Drgoňa et al. [23] applied a white-box MPC in a smart HVAC system for an office building served by a ground-source heat pump. They demonstrated a 53.5 % reduction in initial energy use and a 36.9 % improvement in thermal comfort compared to rule-based controls. Toub et al. [24] implemented MPC in a solar-driven smart HVAC system, resulting in 37 % energy savings and 70 % cost reductions compared to a rule-based controller. Lee et al. [25] demonstrated that MPC for a building cooling system utilizing chilled water from a TES results in an 8.3 % reduction in costs compared to rule-based logic, particularly under conditions of dynamic pricing and varying occupancy. Hilliard et al. [26] demonstrated the implementation of MPC in a community college building utilizing a borehole TES-driven heat pump, resulting in a 9 % reduction in HVAC energy use relative to rule-based controls. Sun et al. [27] introduced a new MPC algorithm combined with guided policy search for air-source heat pump systems, reducing temperature fluctuations to 0.8 °C and enhancing energy efficiency by more than 12 %. Taylor et al. [28] implemented MPC in an HTC and LTH utilizing water-source heat pumps, demonstrating enhanced performance from techno-economic perspectives in smart operations relative to rule-based approaches, despite higher initial investment costs. Razmara et al. [29] employed MPC for Building-to-Grid systems that incorporate photovoltaic and energy storage systems, achieving a 28 % reduction in energy costs and over a 70 % decrease in peak load ramps relative to traditional controllers. Most of these studies were conducted in simulation environments or isolated subsystems, frequently under the assumption of

perfect forecasts and ideal conditions. Applications of MPC in conjunction with borehole TES and hybrid LTH–HTC systems are currently scarce. Furthermore, the scalability of MPC in the context of realistic uncertainties and pricing structures has not been thoroughly assessed.

Although a growing body of literature highlights the advantages of LTH-HTC systems and their integration with borehole TES, their full potential remains underexploited without advanced control strategies. Previous studies have shown that such systems can reduce building energy use by 10–20 % and operational costs by up to 10 %. However, most have relied on simplified models or short-term simulations that do not capture real-world operations or long-term seasonal dynamics. Moreover, conventional rule-based control approaches cannot anticipate market fluctuations, storage imbalances, or varying thermal loads, leaving an estimated 10–15 % of annual energy-savings potential unrealized. Few studies have examined these systems within already optimized, ultra-efficient buildings, where the incremental benefits of advanced predictive control become both technically and economically significant. To address these gaps, this study investigates an ultra-efficient commercial building in Uppsala, Sweden, which integrates HTC and LTH distribution systems with borehole TES, ground-source heat pumps, active chilled beams, and smart ventilation lines for combined heating and cooling. The analysis employs model predictive control (MPC) as a dynamic, forecast-driven optimization layer that operates above existing control systems. The novelty of this work lies not in each individual technology, but in demonstrating how MPC can serve as the final optimization layer, making an already high-performing building even smarter by reducing operational costs and emissions while maintaining comfort (see Fig. 1).

An optimal MPC framework is developed utilizing a Differential Evolution algorithm, addressing a constrained optimization problem at each time step to minimize a market-responsive cost function. A benchmarking analysis is performed against a conventional rule-based control. In addition, while benchmarking and price-responsive operation have been considered in previous MPC studies, this work distinguishes itself by integrating them with multi-layer penalties and robustness testing in a validated, real-world building case that is already operating optimally. The impact of varying prediction horizons (24 h versus 48 h) is systematically evaluated, providing insights into the trade-offs between responsiveness and foresight in predictive building control.

The main contributions of this study are summarized as follows:

- Adaptive MPC framework for advanced thermal systems: This study presents a multi-layered MPC strategy for a heat pump system integrated with borehole TES, facilitating dynamic adaptation to hourly fluctuations in electricity prices, in contrast to the seasonally fixed district heating tariffs in Sweden.
- Comprehensive optimization approach: The MPC cost function incorporates both immediate operational objectives (electricity and heating costs) and system-level impact penalties, including CO_2 emissions, TES preservation, comfort deviations, and peak-load constraints, allowing for more holistic optimization compared to conventional control methods.
- Robustness under uncertainty: The performance of the MPC framework is evaluated through Monte Carlo simulations under forecast uncertainty, confirming its robustness across multiple seasonal scenarios and input perturbations while maintaining system stability and thermal comfort.

This paper is structured as follows. Section 2 presents the case study, detailing the building characteristics and the configuration of the proposed advanced thermal system. Section 3 outlines the methodological framework, including the simulation setup, formulation of the MPC strategy, and sensitivity analysis addressing forecast uncertainty. Section 4 addresses the validation of the model using measured data to confirm the reliability of the simulation framework. Section 5 presents the primary findings, including the sensitivity assessment and a comparative analysis of the rule-based and MPC strategies. Section 6 presents the conclusions and recommendations, followed by the Appendix and references.

2. Case study

This section presents the case study of the Juvelen building, located in Uppsala, central Sweden, an ultra-efficient commercial property that incorporates a modern HVAC system, integrating low-temperature

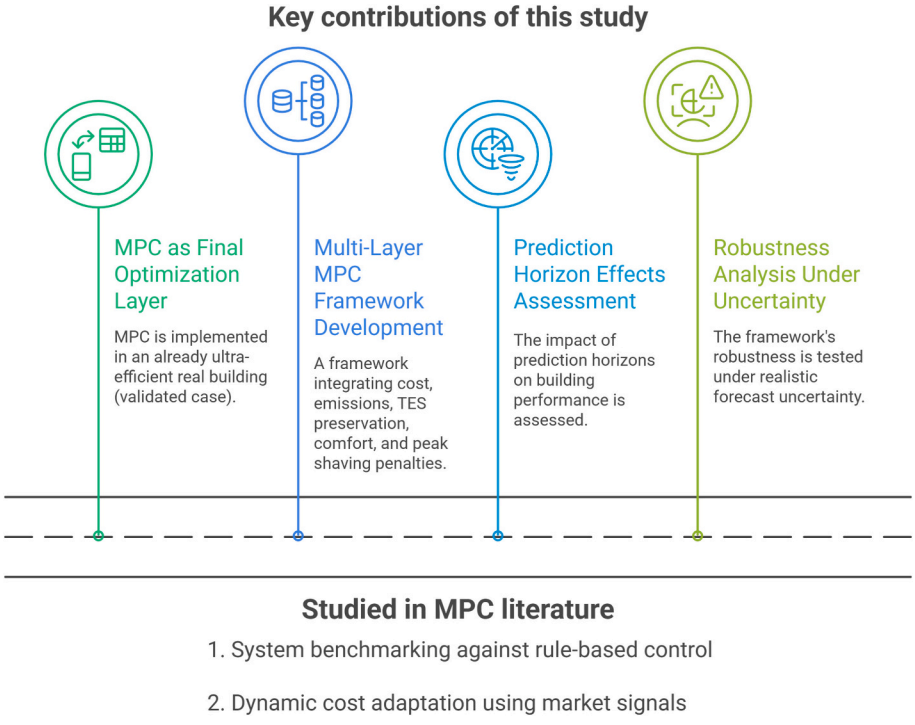


Fig. 1. Overview of the study's scope and contributions.

heating, high-temperature cooling, and borehole thermal energy storage with ground-source heat pumps. It outlines the building characteristics and system configuration that facilitate the implementation of predictive control.

2.1. Building

Juvelen, a prominent commercial property situated beside Uppsala Central Station in central Sweden, is examined in this research [30]. Completed in 2019, the six-story building, with a gross floor area of roughly 10,000 m^2, has been outfitted with adaptable office suites, conference facilities, and a ground-floor restaurant and is regarded as a regional hub for innovation and sustainability.

Juvelen has been certified LEED Platinum and internally classified as a "Dark Green" project, the highest tier of sustainability in the environmental system [31]. Energy efficiency and contemporary Scandinavian aesthetics are conveyed by a steel-and-glass façade reminiscent of a cut gemstone, while transparency, generous daylighting, and spatial efficiency are ensured by a layout that accommodates diverse tenant requirements through flexible interior zones. The building's robust envelope, substantial thermal mass, and effective load-shifting strategies facilitate reduced reliance on fossil fuels. Through these features, Juvelen is operated as a climate-resilient, ultra-efficient facility, offering insights into the future of commercial real estate in cold-climate regions.

2.2. LTH and HTC system

Juvelen's energy system is fundamentally based on the so-called Deep Green Cooling innovation, which utilizes deep-borehole thermal energy storage through a closed-loop brine circuit with boreholes that are 250 m deep. This system circulates a water or water-glycol mixture through a vertical borefield to leverage the stable underground temperature, which ranges from 8 °C to 12 °C during a normal year in Uppsala (see Fig. 2). This configuration enables efficient thermal exchange between the building and the ground, providing both heating and cooling capabilities through smart management.

2.2.1. Energy system configuration

The energy system configuration is based on a hybridization of ground source and district heating (DH) to minimize purchased energy and peak loads. Three independent air handling units (AHUs) serve the building: two are dedicated to office zones, and one serves the ground-floor restaurant. Each AHU includes exhaust-air heat recovery with a rotary heat exchanger, supply and extract fans, and two water-based coils: a heating coil connected to the DH network and a cooling coil coupled with the borehole TES via a plate heat exchanger for free cooling during summer. AHUs also feature heat recovery units (in offices) and a run-around heat exchanger (in the restaurant), which recover waste heat from the exhaust stream, thereby reducing external energy demand and enhancing indoor comfort.

2.2.2. Heating system

The heating system operates as a hybrid configuration that integrates two 40 kW GSHPs assisted by the DH substation. During the heating season, the GSHPs extract low-grade thermal energy (≈ 8 °C) from the boreholes and upgrade it to useful supply temperatures of 30–50 °C, characteristic of nowadays LTH operation. The upgraded heat is distributed through hydronic radiators connected to a buffer tank, while the district heating exchangers provide the additional heat when needed to cover peak space heating demand. Two other DH heat exchangers serve different purposes: one for domestic hot water production and one for post-heating in the AHUs. This hybridization ensures operational flexibility and cost optimization by prioritizing the most economical and low-emission source at each time step.

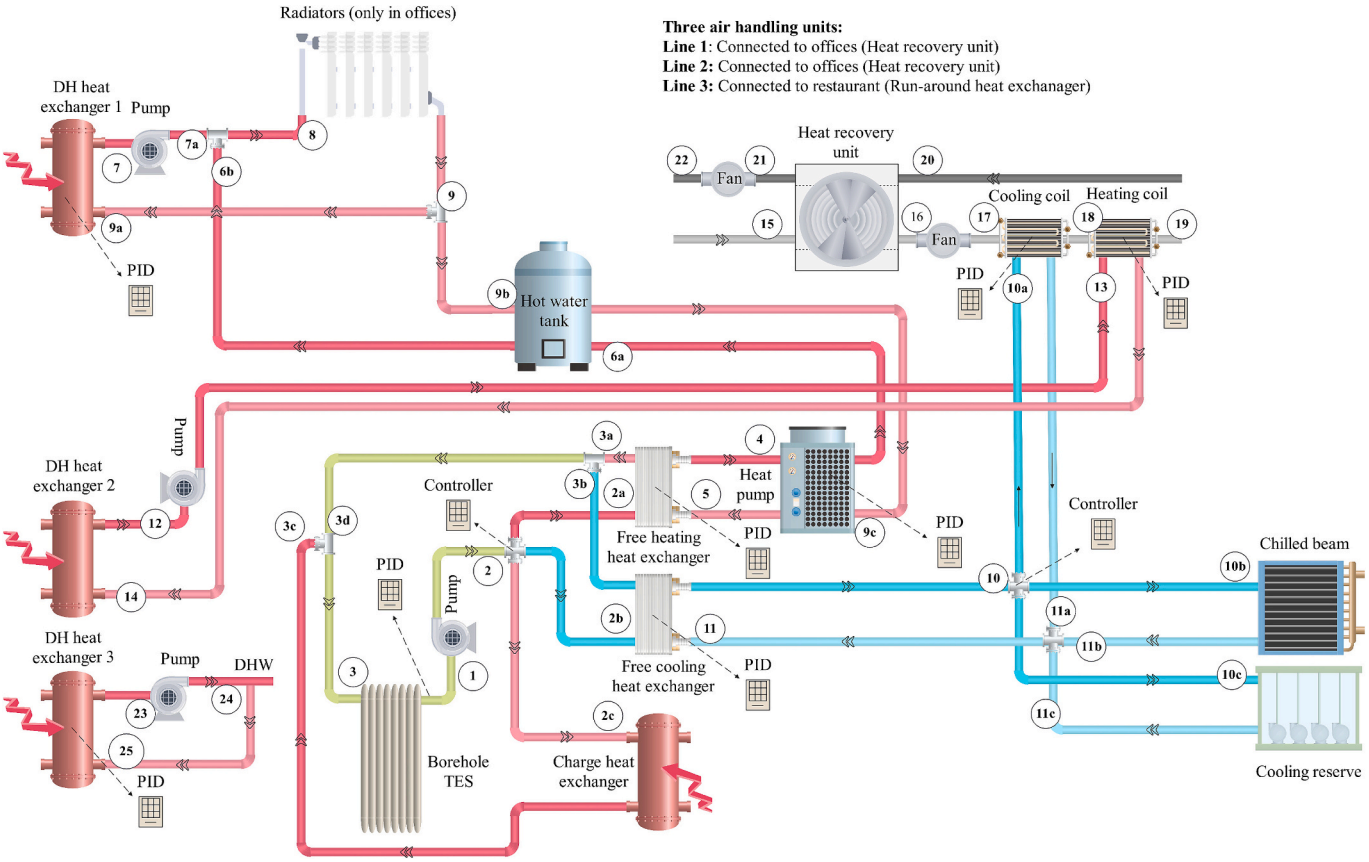


Fig. 2. The schematic representation of the proposed LTH and HTC system based on ground source heat pumps and DH network.

2.2.3. Cooling system

The cooling system is based on HTC principles (cooling supply of >16 °C). The harvested low-grade ground thermal energy is distributed through active chilled beams and post-cooling coils in the building. It enables a compression-free, passive cooling without the need for conventional electric-driven chillers. This strategy significantly reduces electricity usage for active refrigeration while maintaining comfortable indoor conditions. The free-cooling operation is achieved through the same borehole TES loop that exchanges heat with the ambient ground via the brine circuit.

2.2.4. Seasonal storage and maintenance strategy

A key feature of the present configuration is the seasonal storage and maintenance strategy implemented to preserve long-term borehole thermal balance. During summer and early autumn, typically in September when DH seasonal costs are low in Sweden, heat is reinjected into the borehole field via a dedicated charge heat exchanger connected to the DH substation. This recharging of the boreholes compensates for the winter heat extraction, thereby mitigating the gradual long-term cooling of the boreholes. For the proposed seasonal management, a PID controller monitors the average borehole temperature. The heat is injected when the storage temperature drops below the average temperature at the beginning of the same year.

2.2.5. Existing control logic

In the existing system, the control logic is based on a rule-based framework that utilizes ON–OFF and PID controllers. The ON–OFF logic controls the activation of the compressor in the heat pump and the circulation pump connected to the chilled beam unit. PID controllers regulate (1) the heating water circulating in the hydronic radiator to maintain the desired indoor thermal comfort and setpoint temperature of supply hotwater, (2) the mass flowrate of water entering the post-cooling coil in the AHUs to achieve supply-air temperature setpoint at summer, (3) the mass flowrate of heating water entering the post-heating coil in the AHUs to achieve supply-air temperature setpoint at winter, (4) borehole circulation pumps and heat exchangers based on predefined temperature thresholds for the supply and return brine temperatures, (5) the borehole brine flowrate to prevent excessive extraction during winter, and (6) the hot water flowrate entering from heat pump condenser into the tank. While effective for basic regulation, such feedback-only control cannot anticipate future dynamics or optimize operations under variable tariffs. To overcome these limitations, this study introduces an MPC that coordinates the heat pump operation, TES charging, and load shifting decisions. The MPC leverages short-term forecasts of weather and electricity prices to minimize a multi-layer cost function that considers comfort, CO_2 emissions, and peak penalties, as well as operational heating and electricity costs. Fig. 2 serves as a simplified schematic designed to depict the primary thermal and hydraulic connections within the system. The placement and function of specific components, such as 3-way valves and pumps, may vary. For example, 3-way valves are generally utilized for flow mixing rather than flow division.

3. Method

This section presents the methodological framework for simulating, optimizing, and assessing the performance of the advanced LTH–HTC system integrated with borehole TES in the Juvelen commercial building. The methodology comprises three primary subsections. Section 3.1 outlines the detailed modeling of building, HVAC, and TES components utilizing the TRNSYS simulation environment and the co-simulation method established for integration with optimization routines. Section 3.2 presents the MPC formulation, detailing the development of the cost function that incorporates electricity and district heating costs, CO_2 emissions, peak shaving, borehole TES protection, and thermal comfort. Section 3.3 analyzes the framework for sensitivity analysis in the context of forecast uncertainty, evaluating the robustness of the MPC strategy through stochastic simulations that incorporate variations in weather and electricity price profiles.

3.1. System modeling and co-simulation framework

The TRNSYS 18 simulation environment, a popular transient systems simulation tool for modeling the dynamics of building and energy systems, is utilized to model each component of the proposed system. The building components are all represented as TRNSYS Types, programmed in Fortran, and assembled into dynamic-link libraries (DLLs). At each simulation time step, these parts exchange data through inputs and outputs in a modular approach. Additional information regarding the modeling setup is available in the Appendix. Table A1 in the Appendix summarizes the TRNSYS component types and their respective functionalities, while Table A2 details the essential thermo-physical and heat transfer parameters used in the simulations. Fig. A1 presents the hourly volumetric flow rate schedule for air entering each ventilation line, corresponding to zone-specific occupancy and operational schedules.

TRNSYS offers a robust platform for thermophysical modeling; however, it is unsuitable for direct application in optimization-based iterative control methods like MPC. This limitation is attributed to two primary factors: (1) the architecture of the TRNSYS simulation database and (2) its data management structure, which depends on sequential text-based file reading and writing. The characteristics of TRNSYS limit its ability to manage large-scale optimization routines that necessitate frequent state resets and high-speed iterative loops. Therefore, a co-simulation framework is developed to establish the model predictive control.

Alternative co-simulation environments such as Modelica with JModelica.org, EnergyPlus with functional mock-up units, or MATLAB's MPC Toolbox also offer dynamic modeling and control capabilities. However, Modelica focuses on symbolic modeling and requires extensive validation of components for HVAC systems. In contrast, TRNSYS, the simulation tool used here, provides a comprehensive library of experimentally validated building and system components. EnergyPlus is suitable for building-level control but less adaptable for detailed component-level thermal systems. MATLAB's MPC Toolbox is computationally efficient for linearized systems but less suited to nonlinear, high-dimensional HVAC dynamics. Therefore, TRNSYS manages the physical system simulation in this setup, and a separate C++ layer performs the MPC optimization. This hybrid setup enables the retention of detailed, validated component models from TRNSYS, thereby eliminating the need for revalidation, while facilitating high-performance, iterative control via external software. The MPC routine is implemented in C++ and integrated with the PyGMO (Python Parallel Global Multi-objective Optimizer) library, enabling real-time optimization based on system forecasts and constraints. PyGMO is selected due to several key advantages: This approach facilitates parallel global optimization via an island model, promoting effective convergence within an extensive solution space. Secondly, it offers detailed control for each variable, crucial for handling time-varying decision variables in MPC.

Additionally, it provides a variety of evolutionary and deterministic algorithms suitable for constrained, high-dimensional, and non-convex problems. Finally, it integrates easily with C++ codebases and supports customized solver strategies and constraint formulations. Such partitioned co-simulation frameworks have been increasingly applied in building energy optimization and predictive control studies, where physics-based modeling is coupled with external solvers to strike a balance between accuracy and computational efficiency [13,20,22]. This TRNSYS–PyGMO co-simulation framework demonstrates a methodological contribution of this work, enabling non-linear, non-convex MPC to be applied directly to high-fidelity physical models without linearization. This concept bridges the gap between theoretical optimization design and real building applications.

Fig. 3 illustrates the flowchart of the co-simulation framework

developed for the optimal control of the building energy system. TRNSYS facilitates the physics-based simulation of building components, HVAC subsystems, borehole thermal energy storage, and system benchmarking through Fortran-based dynamic link libraries. System states and forecasts, such as weather and electricity prices, are input into a custom-built MPC framework developed in C++ at each time step. This framework incorporates the control logic, prediction model, cost function, and constraints. The optimization problem is addressed using the PyGMO library, which serves as a parallel solver engine that utilizes a Differential Evolution algorithm. The optimized control set-points are provided to TRNSYS for real-time implementation, thereby ensuring synchronization and closed-loop performance.

It should be emphasized that PyGMO is not considered an MPC tool or framework. The present study independently devises the complete MPC architecture, including the prediction model, cost-function formulation, and constraint handling, in C++. PyGMO is employed solely as the optimization engine within each control step of the MPC loop, its role being confined to solving the nonlinear optimization problem generated at every time step. This distinction must be made clear to prevent any misunderstanding that PyGMO supplies a ready-made MPC environment; instead, it is utilized as a powerful and flexible solver embedded within a custom-built MPC framework [32,33].

Table 1 indicates the roles and capabilities of TRNSYS, Fortran, and C++/PyGMO in the co-simulation framework [32,33].

Table 1
Roles and limitations of TRNSYS, Fortran, and C++/PyGMO modules in the hybrid co-simulation setup.

Tool	Role in framework	Strength	Limitation
TRNSYS	Physical modeling of the building Thermodynamic modeling of the HVAC components Fulfill techno-economic-environmental modeling	Widely validated Modular component-based structure Detailed thermophysical modeling	Text-based input/output handling Incompatible with iterative optimization loops Non-parallelizable
Fortran (TRNSYS types)	Backend engine for dynamic simulation of system components	Flexible component logic Stable numerical methods	Implementation based on DLLs prevents optimization within loops No built-in support for backup/restore logic
C++ (MPC framework)	Implementation of MPC logic: prediction model, cost function, constraints	Full control over the MPC structure High implementation speed Flexible integration	Custom synchronization with TRNSYS is required Responsible for managing all forecasting and control coordination.
PyGMO (in C++)	Optimization engine used within each MPC step	Extensive algorithm library Parallel optimization via the island model Precise management of variables	Not an MPC framework Cannot interface directly with TRNSYS Needs external MPC formulation

3.2. Model predictive control formulation

This study implements an MPC control aimed at minimizing a comprehensive cost function across a finite prediction horizon, while adhering to system dynamics, physical constraints, and operational requirements. At each time step, the controller predicts future system behavior utilizing a calibrated simulation model and resolves an optimization problem to identify a sequence of control actions that minimizes total cost. The initial step of this optimal sequence is executed, and the procedure is reiterated at the subsequent time step utilizing revised forecasts and state measurements, thereby creating a rolling horizon

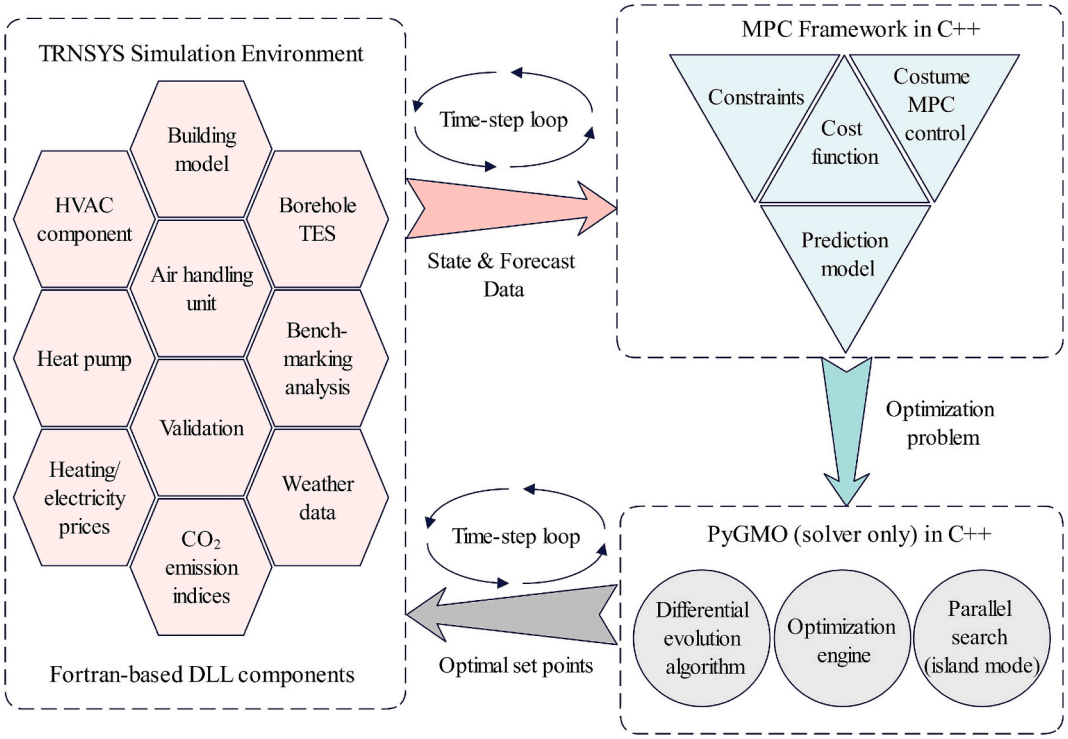


Fig. 3. Flowchart of the co-simulation framework illustrating the interaction between TRNSYS, Fortran, and the C++ – based MPC environment.

control scheme.

The market-responsive operational cost is adopted as the primary objective function, and it integrates diverse economic and environmental metrics. Hourly electricity and seasonal heating prices, CO_2 emissions, system-stress penalties (including peak shaving and thermal-energy storage preservation), and deviations from comfort targets are all incorporated, allowing the MPC to generate cost-effective and sustainable operating decisions in response to dynamic boundary conditions, weather, occupancy, and utility tariffs. The decision variable is the control signal governing the operational state of the heat pump at each time step within the prediction horizon. The primary constraint in the optimization is the temperature limit of the storage tank, which defines the feasible range for charging and discharging actions. The inputs to the MPC include the forecasted electricity spot prices and outdoor air temperature profiles over the prediction horizon, which are used to anticipate system demand and energy cost fluctuations. For this study, the optimization problem has medium dimensionality; therefore, the population size in the Differential Evolution algorithm is five times the number of decision variables. The stopping criterion is defined as the absence of significant fitness improvement, specifically when the improvement in the best objective function value fell below 10^{-4}. The control parameters of the Differential Evolution algorithm include the mutation factor of 0.8 and the crossover rate of 0.9. In the PyGMO implementation, an internal limit of 150 generations was used to prevent excessive runtime, followed by an outer adaptive loop that continued until the 10^{-4} fitness improvement threshold was satisfied. This two-level termination mechanism ensured both computational efficiency and convergence accuracy.

The optimization problem exhibits non-convexity and nonlinearity due to the dynamic interactions within the system and the mixed-integer characteristics of certain control modes. The solution is derived by integrating a differential evolution algorithm into a C++ framework, guaranteeing robustness and effective global search capabilities. This predictive method enables the system to proactively adjust thermal loads, utilize low-tariff periods, and effectively manage thermal storage, resulting in a notable benefit compared to traditional rule-based or reactive control strategies. The MPC routine is integrated with the PyGMO library to address the resulting nonlinear optimization problem, thereby facilitating efficient and parallelized solutions for high-dimensional control variables in real-time. The Differential Evolution algorithm is selected because it is particularly suitable for nonlinear and non-convex optimization problems with multiple interacting constraints, such as those arising in predictive control of hybrid HVAC–TES systems. Differential Evolution offers several advantages over conventional solvers. First, it is derivative-free, avoiding difficulties in handling discontinuities and mixed-integer variables. It provides a global search capability, reducing the risk of convergence to local minima. The parallel island-model implementation in PyGMO improves computational efficiency, making near real-time MPC feasible.

In the present work, a multi-layered objective function (market-responsive operational cost) is developed to evaluate and optimize the performance of the proposed energy system, integrating multiple critical performance indicators into a unified cost formulation. The performance indicators included in the objective function are selected to reflect both practical operational needs and long-term sustainability goals. Market-responsive operational costs are the core indicator because they directly link system operation to fluctuating electricity and heating tariffs. To not only focus on short-sighted operational cost optimization, four additional penalty terms are introduced: CO_2 emissions (environmental performance), borehole TES preservation (long-term thermal source stability), peak shaving (grid and demand-side management/integration), and thermal comfort (occupant well-being). This combination ensures that economic savings are achieved without compromising environmental impact, system degradation, or comfort. This improved cost function incorporates electricity ($C_{electricity}(t)$) and district heating ($C_{heating}(t)$) costs, CO_2 emissions ($C_{emission}(t)$) and peak shaving penalties ($C_{peak}(t)$), borehole TES temperature preservation costs ($C_{TES}(t)$), and deviations from comfort levels penalty ($C_{comfort}(t)$). Although the literature has addressed each term individually, the present approach integrates them into a unified formulation, facilitating a more robust and comprehensive optimization framework as shown below [34–38]:

$$J = \sum_t \left[C_{electricity}(t) + C_{heating}(t) + C_{emission}(t) + C_{peak}(t) + C_{TES}(t) + C_{comfort}(t) \right] \quad (1)$$

$$C_{electricity}(t) = c_{el}(t) \cdot E_{grid}(t) \quad (2)$$

$$C_{heating}(t) = c_{DH}(t) \cdot Q_{DH}(t) \quad (3)$$

$$C_{emission}(t) = \beta \cdot \left(e_{el}(t) \cdot E_{grid}(t) + e_{DH}(t) \cdot Q_{DH}(t) \right) \quad (4)$$

$$C_{peak}(t) = \gamma \cdot P_{HP}(t)^2 \quad (5)$$

$$C_{TES}(t) = \delta \cdot \left(T_{source,min} - T_{TES}(t) \right)^2 \quad (6)$$

$$C_{comfort}(t) = \lambda(t) \cdot (\max(0, |T_{zone}(t) - T_{set}| - \Delta T_{tol}(t)))^2 \quad (7)$$

In Eq. (1), J is the market-responsive operational cost, which is the optimization objective of the MPC framework. In this equation, each term addresses a critical dimension of system operation. In this equation, no explicit weighting coefficients are assigned to individual terms of the cost function. Consequently, the optimization problem remains balanced and unbiased, with all criteria treated equally. While Eq. (2) reflects the electricity costs related to the spot and transmission prices, Eq. (3) indicates the distinct heating costs accounting for the seasonal pricing of heat bought from the providers in Sweden. $c_{el}(t)$, $E_{grid}(t)$, $c_{DH}(t)$, and $Q_{DH}(t)$ are, respectively, the spot+transmission electricity cost in SEK/kWh, total electricity bought from the grid in kWh to run heat pumps and auxiliary components, the seasonal heating costs in SEK/kWh, and the total heat bought from the DH network in kWh. As shown in Fig. 4, the Swedish electricity billing structure comprises four components: spot price (SEK/kWh), transmission tariff (SEK/kWh), power-based charge (SEK/kW per month), and electricity tax (SEK/month). Heating costs comprise two primary components: seasonal energy prices (SEK/kWh) and fixed capacity-based fees (SEK/kW per year). In this study, the market-responsive operational cost formulation exclusively considers the variable energy components: spot and transmission for electricity, and seasonal energy price for district heating. This simplification is warranted because these are the sole cost terms that vary over time and are relevant to the optimization horizon, whereas fixed or capacity-based charges remain static or are determined on monthly or yearly intervals and are not directly affected by short-term operational decisions.

Eq. (4) is related to the CO_2 emissions penalty, calculated as the product of the CO_2 intensity of electricity ($e_{el}(t)$) and district heating ($e_{DH}(t)$), multiplied by the carbon tax (β) of 1.2575 SEK/(kg.CO_2) [39]. The CO_2 intensity of electricity, which is based on the national Swedish electricity mix and the district heating network provided by the local district heating company, can be found in [40,41]. Additionally, Eq. (5) represents the peak shaving penalties, discouraging excessive instantaneous power use from heat pumps ($P_{HP}(t)$), enhancing load shaving and system responsiveness. The present work applies a mild penalty (γ) of 0.015 SEK/kW^2 to promote better grid integration. Eq. (6) shows the borehole TES protection costs, defined to prevent long-term degradation of boreholes by penalizing depletion below a minimum ground temperature ($T_{source,\ min} = 0\ ^\circ C$), with δ of 101.6 SEK/$^\circ$C^2. The last, but not least, important term in the cost function formula is the comfort penalty (Eq. (7)), an indicator that encourages thermal comfort within a flexible tolerance band, dynamically adjusted based on electricity prices to trade off comfort and cost. The tolerance band ($\Delta T_{tol}(t)$) and weight ($\lambda(t)$) are

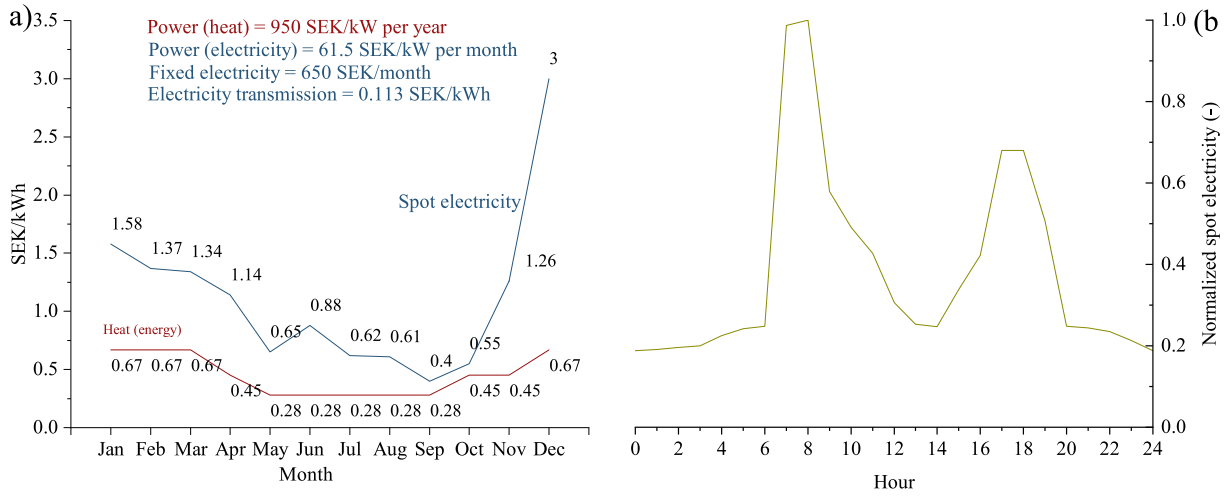


Fig. 4. (a) Monthly distribution of heat and electricity costs and fixed, transmission, and power charges, and (b) normalized hourly variation of spot electricity prices in Uppsala, Sweden [42,43].

calculated using normalized prices ($p_{norm}(t)$), facilitating soft, price-responsive comfort enforcement:

$$\Delta T_{tol}(t) = \Delta T_{max} - p_{norm}(t)\cdot(\Delta T_{max} - \Delta T_{min}) \tag{8}$$

$$\lambda(t) = \lambda_{min} - p_{norm}(t)\cdot(\lambda_{max} - \lambda_{min}) \tag{9}$$

In the present study, ΔT_{max}, ΔT_{min}, λ_{max}, and λ_{min} are 2 °C, 0.5 °C, 10.06 SEK/°C^2, and 1.006 SEK/°C^2, respectively. The environmental and economic indicators incorporated into the MPC cost function are determined according to national policy instruments, operational relevance, and relevance to Swedish market conditions. On the economic side, only the variable components of electricity tariffs (spot and transmission) and district heating prices (seasonal energy fees) are included, since these terms fluctuate on the timescales relevant to MPC. Fixed capacity or tax components, which remain constant on a monthly or annual basis, are excluded because short-term operational changes cannot influence them. The environmental aspect involves the CO_2 emission penalty, which corresponds to the Swedish national carbon tax of 1.2575 SEK/(kg·CO_2), imposed on the carbon intensities associated with electricity and district heating. This formulation is consistent with actual cost structures and emission accounting, guaranteeing that optimization outcomes are relevant to both the market and policy frameworks.

In the MPC formulation, the prediction model is represented by the TRNSYS simulation environment, which has been validated against measured building data (see Section 4). Instead of simplified state–space equations, the validated TRNSYS model forecasts the dynamic evolution of indoor thermal states, heat pump operation, TES charge/discharge, and system energy demands based on weather, occupancy, and control inputs. At each time step, the C++/PyGMO optimizer generates candidate control trajectories, which TRNSYS evaluates; the optimizer then selects the trajectory that minimizes the cost function defined in Eqs. (1)–(7). This co-simulation architecture allows MPC to exploit a high-fidelity prediction model without re-deriving reduced-order equations. However, the proposed MPC framework is associated with various limitations that must be considered. For instance, convergence uncertainty could result in high-dimensional non-convex optimization problems. Additionally, uncertainty in forecasts of electricity prices or meteorological data can lead to long-term bias and the accumulation of errors in forecast-driven control projections. Third, in contrast to most existing MPC frameworks, the present formula does not consider weighting coefficients, as all cost elements are normalized and regarded as identical. Finally, for larger systems or shorter sampling intervals, computational efficiency could be enhanced through reduced-order modeling, online linearization, or anticipatory pre-solving strategies.

3.3. Sensitivity analysis under forecast uncertainty

The performance of MPC is fundamentally reliant on the precision of future forecasts. A comprehensive sensitivity analysis is conducted to assess the robustness of the proposed MPC strategy under realistic uncertainties by introducing stochastic perturbations into the forecast data. Uncertainties are specifically applied to two critical exogenous inputs: outdoor air temperature and electricity spot prices, which directly depend on building thermal loads and energy operational costs.

In the present study, a strategic sampling approach integrated with Monte Carlo scenarios is applied over four representative weeks, encompassing cold winter peaks, shoulder-season transitions, and periods of reduced thermal demand, to capture the varying dynamics of heating, cooling, and transitional periods. For each week, 25 Monte Carlo scenarios are generated utilizing Gaussian noise distributions centered on the original forecast values. The ranges for these perturbations are not arbitrary; they are selected based on historical forecast error statistics for Sweden's SE3 bidding zone. In particular, short-term outdoor temperature forecasts typically deviate by about ±1 °C, and hourly spot electricity prices by ±0.03 SEK/kWh. To represent this realistically, Gaussian noise distributions with zero mean and these standard deviations are applied to generate perturbed scenarios. The forecasts for outdoor temperature and electricity spot price using randomized Gaussian perturbations are written as follows:

$$T^{(i)}_{forecast}(t) = T_{true}(t) + N\left(0, \sigma_T^2\right) \tag{10}$$

$$c^{(i)}_{el,forecast}(t) = c_{el,true}(t) + N\left(0, \sigma_c^2\right) \tag{11}$$

Where σ_T equals 1 °C, representing standard deviations based on historical forecast errors and σ_c is 0.03 SEK/kWh, aligned with typical hourly spot price volatility in Sweden's middle/central (SE3 bidding) zone. In each forecast, $N(0, \sigma^2)$ represents a normal distribution with zero mean and specified standard deviation. Each scenario is simulated independently utilizing the same MPC framework, and the results are compared against a baseline case featuring perfect forecasts. Key performance indicators, including total market-responsive operational cost and savings, are utilized to quantify sensitivity. The sensitivity analysis framework consists of three essential steps, as depicted in Fig. 5. In Step 1, uncertainty is introduced by generating 25 Monte Carlo scenarios for outdoor temperature and electricity spot price signals. Gaussian noise is used to model realistic forecasting inaccuracies, corresponding to standard deviations of ±1 °C for outdoor temperature and ± 0.03 SEK/kWh for price variation, respectively. Variations in noisy inputs act to evaluate the robustness of controllers. Step 2, termed uncertainty

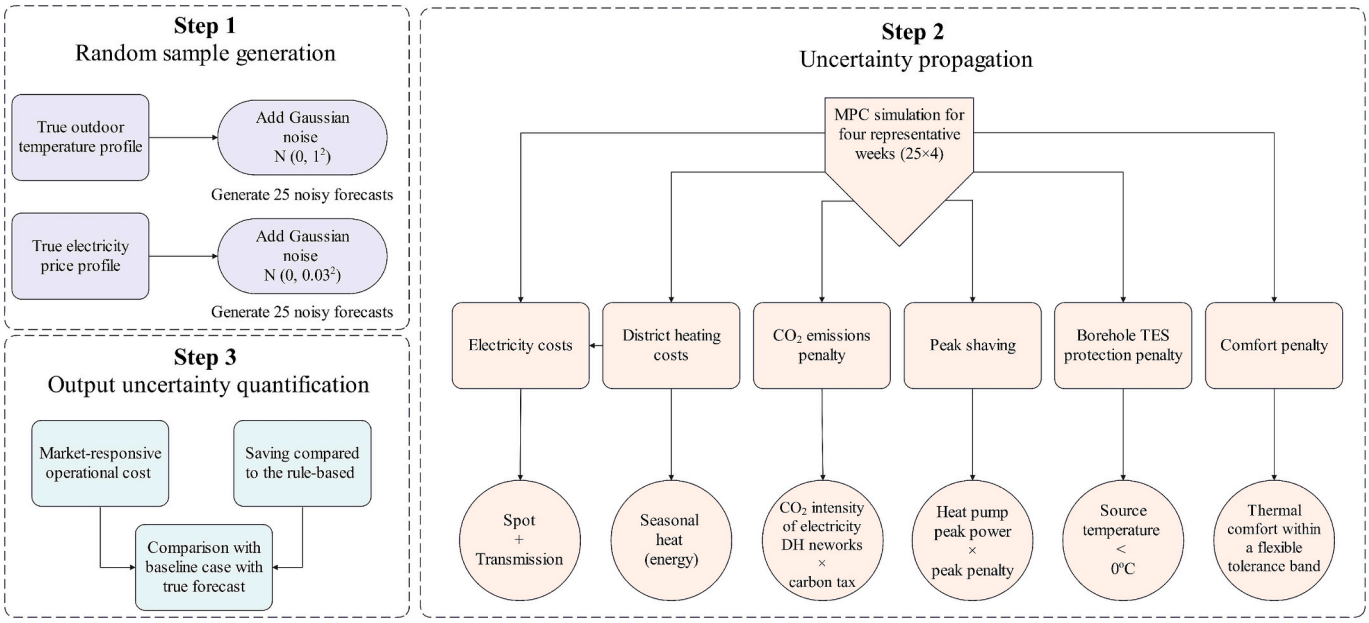


Fig. 5. Flowchart of the stochastic sensitivity analysis framework.

Propagation, entails simulating the MPC strategy across four representative weeks (heating-dominated, less heating-dominated, and two transitional periods) for each of the perturbed inputs. Each simulation incorporates all components of the cost function, electricity and district heating costs, CO_2 emission penalties, peak shaving, TES protection, and comfort violation penalties, facilitating a thorough evaluation amidst input variability. In Step 3, a statistical analysis is conducted to examine the distribution of performance indicators. The outputs consist of the market-responsive operational costs and relative savings compared to the rule-based baseline, as well as their deviations from the ideal (perfect forecast) scenario.

4. Validation

The TRNSYS-modeled building thermal behavior is validated in Fig. 6 by comparing the daily heating and cooling loads that are simulated and measured. The validation data were obtained from the Tekla Building Management System (BMS) of the Juvelen building, which continuously records hourly measured thermal energy demand (kWh) from 2021 to 2023. The dataset spans three full years, encompassing both heating and cooling seasons, ensuring that the model captures interannual variability and typical operational conditions. The operator-provided data are plotted on the vertical axis and compared against TRNSYS-simulated values on the horizontal axis. The scatter plots demonstrate the model's predictive power in both temperature domains. With a root mean square error (RMSE) of 353.6 kWh and a coefficient of determination (R^2) of 0.91, the simulated findings for heating closely match the real values.

On the other hand, the cooling model achieves an R^2 of 0.98 and a much smaller RMSE of 45.3 kWh, showing almost perfect alignment with the measurement. The building's steady internal gains and reliable passive cooling system are likely the reasons for this tight clustering, which demonstrates outstanding model performance during cooling periods. These findings support the reliability and dependability of the TRNSYS building model in simulating actual operational dynamics during the heating and cooling seasons. The higher correlation achieved for cooling reflects the relatively stable indoor gains and consistent passive cooling performance, where both internal and environmental conditions remain steady throughout operation. In contrast, the lower heating accuracy is mainly due to the uncertainty in occupancy-driven

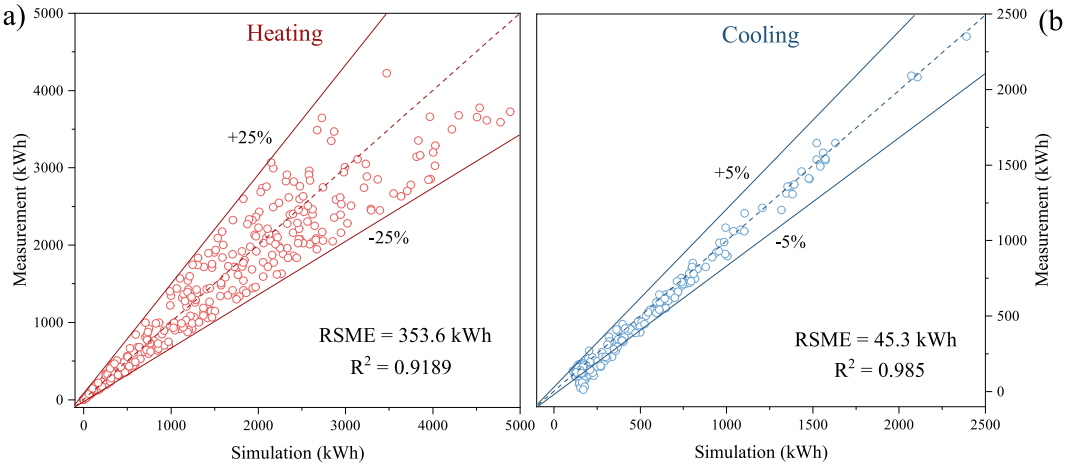


Fig. 6. Validation of simulated heating and cooling loads against measured data from the Tekla BMS of the Juvelen building. (a) Heating and (b) Cooling. All values are displayed at a daily resolution, with axes in kWh. The 1:1 reference line and ± 25 % (heating) and ± 5 % (cooling) bounds are indicated [30,44].

internal gains, especially within the office and restaurant zones, where user presence and equipment loads vary significantly. Consequently, the simulation utilizes standardized occupancy schedules, which may not accurately capture day-to-day variations in real-world operations. Minor differences may also stem from extremely cold hours during the wintertime in practice, simplification of the PID control loops used for radiator and post-heating coils in TRNSYS, and from uncertainties in the borehole TES heat exchange during low-load conditions.

While the validation focused on heating and cooling demand, indoor thermal conditions are not directly compared with measurements because continuous zone-level monitoring data were unavailable. However, the TRNSYS building model is developed from as-built envelope and HVAC specifications, ensuring that simulated indoor conditions remain within realistic comfort bounds. Importantly, in the MPC formulation, the prediction model is not a simplified surrogate but the validated TRNSYS simulation itself. Therefore, the agreement shown in Fig. 6 between TRNSYS and measured heating/cooling demand also constitutes the validation of the prediction model used by the MPC.

Fig. 7 illustrates a heatmap depicting the average relative absolute error between the simulated and measured hourly heat exchange with the borehole thermal energy storage system, represented as a percentage and organized by month and hour of the day. This analysis, which utilizes detailed comparisons with the building management system (Tekla), facilitates the identification of temporal patterns in model accuracy across seasonal and diurnal cycles. The relative error typically varies from 5 % to 14 %, with greater deviations primarily observed during early morning and evening hours in the spring and autumn months, specifically in March, May, and October. The increased errors correspond with swift changes in thermal loads, where prediction difficulties are exacerbated by fluctuating outdoor conditions and variations in occupancy behaviors. In contrast, the summer months (June–August) consistently demonstrate low error levels, frequently below 6 %, indicating a strong correlation between the model and measurements during cooling operations and stable environmental conditions. The figure illustrates the reliability of the TES model, as the relative errors are moderate and evenly distributed throughout the year.

5. Results and discussion

The results are structured to initially demonstrate the robustness of the MPC framework, followed by a description of its performance outcomes. The analysis is catagorized as follows: (1) assessment of forecast uncertainty scenarios via Monte Carlo simulations including Gaussian perturbations of weather and market forecasts, (2) comparison of the performance of Model Predictive Control against traditional rule-based control, (3) analysis of prediction horizons (24 h versus 48 h), and (4) assessment of long-term financial/environmental implications. The decision to start with forecast uncertainty is intentional: by demonstrating that MPC maintains stability under realistic perturbations, confidence is established in the subsequent performance comparisons and long-term evaluations.

5.1. Sensitivity analysis

MPC relies heavily on accurate predictions of external variables, including outdoor temperature and spot electricity prices, to facilitate cost-optimized decision-making. These forecasts inherently contain uncertainties that may undermine control performance in practical

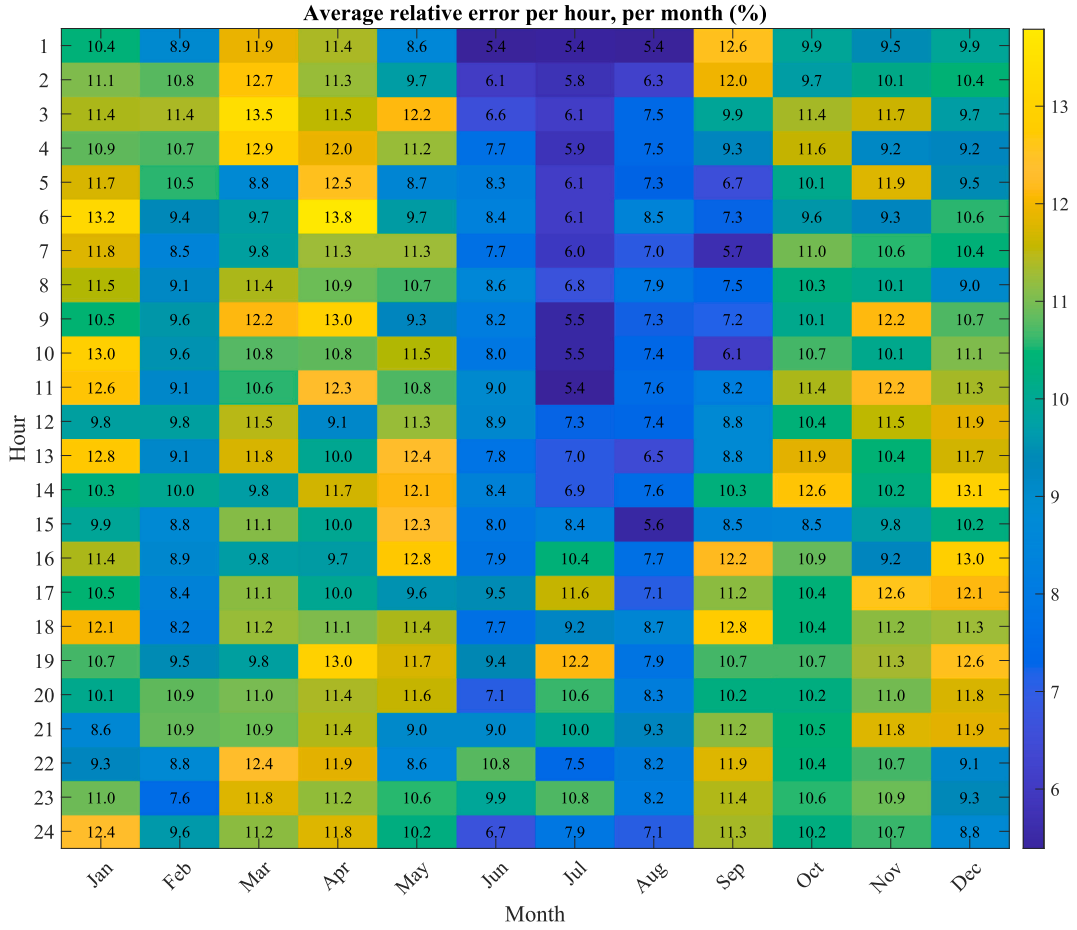


Fig. 7. Hourly–monthly heat-transfer validation of the borehole TES: average relative error (%) per hour and per month obtained from TRNSYS simulations compared with Tekla BMS measurements for the Juvelen building. The color bar quantifies deviation magnitude (5 %–14 %) [30].

applications. This subsection presents a sensitivity analysis to evaluate the robustness of the proposed MPC framework under uncertainty. For this purpose, controlled, zero-mean Gaussian noises are introduced to the temperature and price forecasts to simulate realistic prediction errors and assess their impact on the system's performance. This approach is not applied throughout the entire year due to its computational intensity; instead, four representative weeks are selected to represent various operating regimes, including a cold winter peak, shoulder-season transitions, and periods of low thermal demand. For each selected week, 25 Monte Carlo simulations are performed using randomized input noise to quantify the value and possible savings in market-responsive operational costs compared to the idealized perfect-forecast scenario (base case). Before presenting the sensitivity results, Figs. A1 and A2 in the Appendix illustrate the statistical distribution of the introduced forecast errors in hourly outdoor temperature and spot electricity prices, represented as zero-mean Gaussian noise with a standard deviation of 1 °C and 0.03 SEK/kWh, over a 24-h time horizon.

Fig. 8 illustrates the sensitivity of the MPC controller to forecast uncertainties by depicting the evolution of market-responsive operational cost and cost savings as the noise level in temperature and electricity price forecasts increases. The horizontal axis denotes various standard deviations (σ) of introduced forecast error, extending from perfect forecasts (σ = 0) to scenarios with significant errors (σ = 2.5). The bar plots (left axis) represent the total operational cost under MPC for each noise level. In contrast, the line plot (right axis) illustrates the corresponding savings relative to the fixed baseline of 1802 SEK under rule-based control for the same week.

The figure illustrates that, in ideal forecasting scenarios, MPC yields the highest savings. Control decisions increasingly diverge from actual price and weather conditions as forecast errors escalate. This reduces savings and increases operational costs, indicating lost opportunities for effective heat pump deployment and the utilization of thermal storage. At elevated error levels (σ ≥ 2), the predictive advantage of the MPC diminishes significantly, resulting in performance that closely aligns with the rule-based controllers. This outcome highlights a significant challenge in predictive control: its effectiveness relies not only on system flexibility and optimization logic but also on the reliability of input forecasts. In Nordic energy systems such as Uppsala's, characterized by price volatility and seasonal demand fluctuations, it is essential to integrate forecast-aware control with effective strategies to manage uncertainty and maintain economic and environmental advantages.

Fig. 9 presents the hourly dynamics of the market-responsive operational cost during Week 15 for the MPC controller, analyzed under varying levels of forecast uncertainty, compared to the rule-based baseline. The curves illustrate the total hourly cost arising from MPC decisions influenced by forecasts characterized by different levels of Gaussian noise, whereas the rule-based cost operates without prediction, provided for comparison.

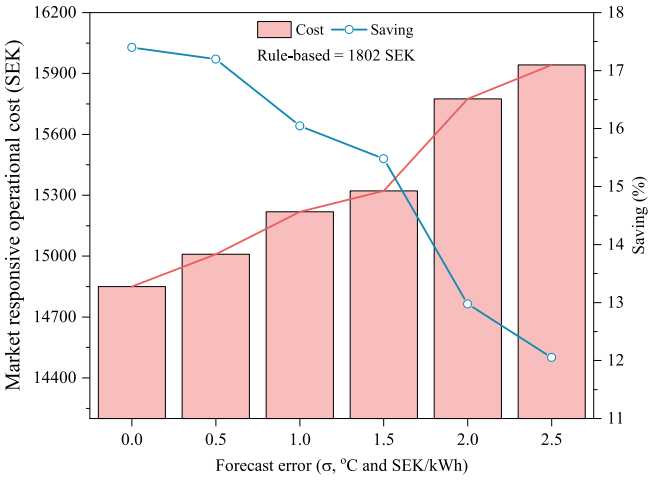


Fig. 8. Impact of forecast error on market-responsive operational cost and savings compared to the rule-based as the reference (Week 15, Horizon = 24 h).

This figure reveals several patterns. The gap between the MPC curves becomes more pronounced during periods of elevated price or load volatility, particularly at midday peaks and early evening ramp-ups. This suggests that forecast uncertainty has a significant impact on critical price-responsive control periods, where even minor inaccuracies in temperature or price predictions can lead to suboptimal system operations. In contrast, during periods of low demand or stable prices, the influence of forecast errors decreases, leading to a closer alignment of the MPC cost curves. This suggests that the controller's decisions exhibit greater resilience when forecast-driven optimization has less of an impact on the system. Although forecast uncertainty introduces operational variability, this figure demonstrates that the MPC exhibits a relatively stable performance envelope and surpasses rule-based control in numerous periods, affirming its effectiveness as a robust, market-responsive strategy, even among realistic forecast errors.

Fig. 10 compares market-responsive operational costs across 25 error scenarios (S1 through S25) alongside the perfect forecast case (base column) for four selected weeks (Weeks 15, 19, 45, and 52). Each subplot represents a week and illustrates the MPC cost under Gaussian noise conditions (temperature: $\epsilon_T \sim N(0, 1\ ^\circ C)$; price $\epsilon_P \sim N(0, 0.03$ SEK/kWh), alongside the rule-based cost depicted as a fixed reference line. According to the figure, during Weeks 15, 19, and 45, the MPC exhibits significant robustness, as most error scenarios result in reduced operational costs compared to the rule-based baseline. This suggests that the MPC's optimization framework continues to function effectively amidst realistic forecast uncertainty, especially during shoulder seasons (spring and autumn), where moderate demand levels and system flexibility enhance adaptation to price signals and thermal storage dynamics. The narrow cost range observed during these weeks suggests that forecast error has a minimal impact on MPC's ability to make cost-effective decisions.

On the other hand, Week 52, occurring in late December, exhibits a divergent trend. The cost spread across various noise scenarios is greater, and in certain instances, the MPC exhibits poorer performance than rule-based control. This change can be ascribed to several interrelated factors: 1) Elevated heating demand resulting from severe winter conditions restricts operational flexibility, 2) heat pumps may be functioning close to their maximum capacity and are using significantly more electricity than during the rest of the year, diminishing the effectiveness of load shifting, and 3) inaccuracies in the prediction of outdoor temperature and spot electricity pricing can lead to misalignment of control actions with the actual needs of the system during high-load peak conditions. This figure illustrates that although MPC typically provides cost-effective and robust control, its advantages may be diminished during periods of high demand and low flexibility, when forecast errors are particularly disruptive.

Fig. 11 presents the distribution of cost savings across 25 forecast error scenarios for each of the four selected weeks, employing violin plots to emphasize the variability of outcomes under uncertainty. The shape of each violin provides significant insights into the distribution and uniformity of savings. In Week 15 (April), the violin plot exhibits symmetry and moderate width at the center, suggesting that most scenarios are concentrated around the mean with restricted variability, indicative of stable and robust MPC performance. Week 19 (May) features a tall, narrow violin shape with flat top and bottom sections, indicating a dense and consistent savings pattern characterized by minimal dispersion, thereby rendering it the most predictable week. In contrast, Week 45 (November) exhibits a wider and more rounded distribution, characterized by fatter tails that indicate a broader range of savings. This suggests that transitional weather conditions increase uncertainty, yet most scenarios continue to provide benefits. Week 52 (December) is notable for its flattened, broad-base violin that extends below zero savings, suggesting that the MPC's performance during the

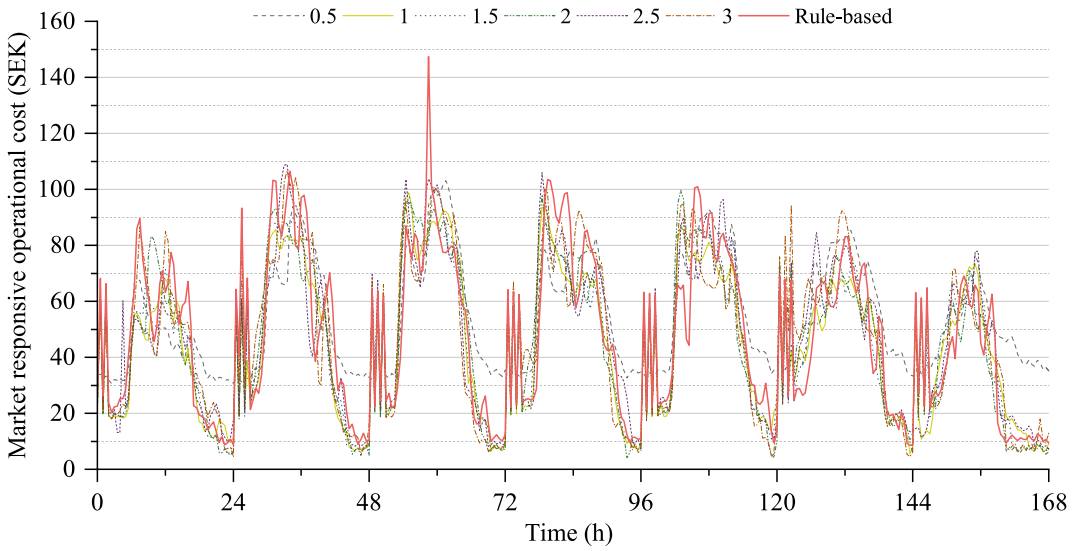


Fig. 9. Hourly changes in market-responsive operational cost under different forecast errors and rule-based (Week 15, Horizon = 24 h).

winter peak load is significantly affected by forecast inaccuracies. The heightened rigidity of the system and increased demand diminish the controller's capacity for flexible response, occasionally resulting in costs that exceed rule-based control. The violin plots illustrate that MPC typically provides consistent benefits; however, its sensitivity to forecast uncertainty fluctuates by season, with winter weeks presenting the highest risk of underperformanceFrom the week-to-week comparison across different seasons, it can also be concluded that there is a strong dependency of MPC effectiveness on available operational flexibility. In winter, the system operates near full capacity to meet high heating loads. It limits the room for optimization and restricts MPC's ability to shift or defer operations. In contrast, in spring and autumn, when heating demand is lower and both heat pumps and TES operate within part-load conditions, the MPC can strategically plan energy use and storage to minimize costs. These shoulder-season periods, therefore, reveal the full potential of predictive control in exploiting price dynamics and storage flexibility.

Table 2 presents a statistical summary of cost savings across various forecast error scenarios for four selected weeks. Week 19, selected from late May, exhibits notable robustness and performance, with a mean saving of 63.4 % and a standard deviation of 0.34 %, indicating consistent and substantial advantages from MPC despite forecast uncertainties. In Week 15, early April, the data indicates a mean savings of 15.9 %, accompanied by a standard deviation of 0.98 %. This suggests a moderate sensitivity to input errors. Week 45, occurring in early November, is notable for its stability, providing moderate savings with a mean of 8.06 % and minimal variation, indicated by a standard deviation of 0.12 %. Conversely, at the end of December, Week 52 indicates a negative mean saving of −6.3 %, demonstrating that MPC performance may be detrimental under specific operating conditions. These results highlight that MPC can provide significant and dependable savings during specific weeks, yet seasonal dynamics and the accuracy of input forecasts influence its effectiveness.

5.2. Comparative assessment of rule-based and MPC with varying prediction horizons

The benefits of implementing model-predictive control (MPC) are evaluated by comparing system performance from a techno-economic and environmental perspective across three control strategies: baseline rule-based control and MPC with 24- and 48-h horizons. The entire operational year is analyzed, with key metrics reported on a monthly, seasonal, and annual basis. Through this comparison, both direct cost savings and broader systemic impacts of predictive control are quantified. The system's responsiveness and operational efficiency are influenced by extending the MPC horizon, so informed decisions can be made under fluctuating electricity prices, ambient temperatures, and internal thermal inertia.

Table 3 presents a comparative analysis of techno-economic performance indicators for the proposed system, utilizing both rule-based control and model MPC, across 24- and 48-h time horizons. The table reports the annual values based on hourly simulations. The introduction of MPC results in a 4.8 % increase in energy-related heating costs over 24 h and a 2.9 % increase over 48 h, while power-related heating costs remain stable. The increase in heat energy costs can be attributed to a strategic shift from electricity to heat during periods of low prices, aimed at mitigating potential penalties, stabilizing long-term operations, or avoiding future electricity spikes. Significant savings are observed in the spot and transmission components of electricity prices. Spot electricity costs decrease by 11.7 % with the 24-h MPC and 9 % with the 48-h horizon, while transmission charges decline by over 14 % and 11 %. The fixed and power-based charges are unchanged, as they are dictated by pre-established capacity contracts that rely on annual peak heat and electricity demands. The peak values remain constant regardless of real-time control strategies; thus, the associated charges are not influenced by the application of MPC.

The table further indicates that while the total heating (energy + power) price increases by approximately 2.2 % and 1.3 % over 24-h and 48-h horizons, the electricity-related costs, including spot, power-based, fixed, and transmission fees, decrease by more than 8 % and 6 %, respectively. This illustrates the MPC's capacity to strategically shift electricity usage to periods with lower market rates, thus reducing vulnerability to fluctuating or peak electricity costs. MPC effectively leverages electricity market dynamics while ensuring heat delivery, demonstrating its enhanced temporal flexibility and cost-optimization capabilities compared to rule-based control. Throughout the year, heating energy use rose by approximately 2 %, whereas electricity consumption declined by about 6 %. The comparison demonstrates the MPC framework's capacity to effectively balance energy sources, transitioning between electricity-driven heat pumps and district heating in response to tariff structures and system conditions. The controller optimizes overall operational cost and efficiency, rather than merely minimizing one energy stream, by showcasing intelligent sector coupling through predictive decision-making.

The table further indicates that a major contribution to cost savings comes from reducing system impact penalties. The MPC significantly

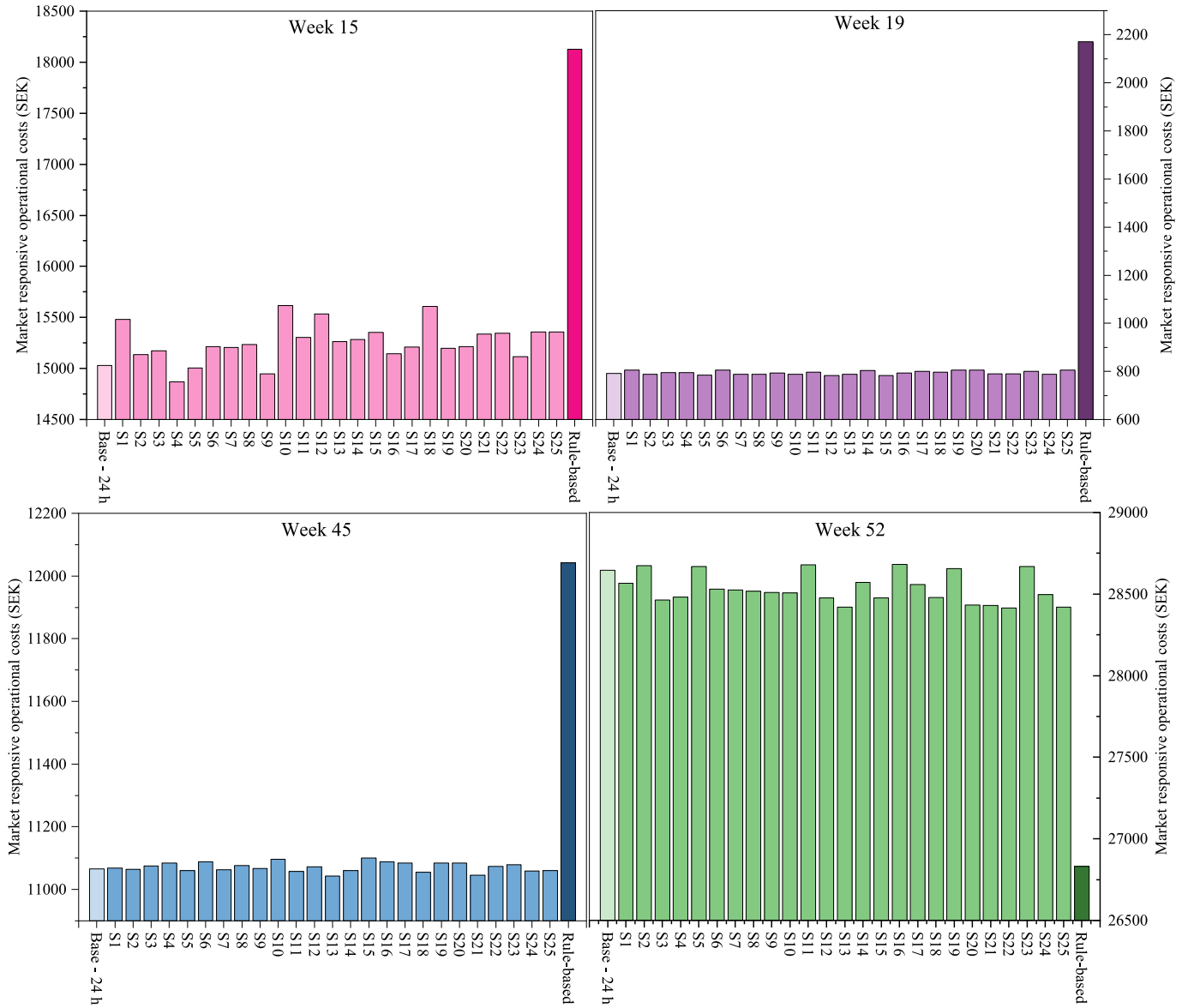


Fig. 10. Market-responsive operational costs under different error scenarios (S1 through S25) and perfect forecast error (base), applying MPC and rule-based control (Selected weeks, horizon = 24 h).

reduces borehole thermal energy storage (TES) preservation costs, decreasing by over 85 % for a 24-h horizon and exceeding 90 % for a 48-h horizon. The significant reduction indicates that the MPC algorithm utilizes TES resources more effectively, managing the charging and discharging of boreholes to satisfy short-term needs while maintaining long-term thermal balance, thus reducing the necessity for preservation actions or penalties. Furthermore, peak shaving penalties, which indicate expenses during periods of high demand, are reduced by approximately 22.6 % and 18.7 % for the 24-h and 48-h MPC scenarios, respectively. This trend demonstrates enhanced load shifting and demand flattening capabilities facilitated by MPC, enabling the system to avoid costly peaks more effectively than a rule-based control system.

Comfort-related penalties, which indicate deviations from desired indoor temperature ranges, are significantly minimized, nearing zero under MPC operation. This demonstrates that MPC reduces operational costs while simultaneously improving indoor thermal comfort through proactive adjustments of system outputs based on predicted conditions. The penalties associated with CO_2 emissions show negligible variation, reflecting a modest increase of 0.7 % for the 24-h horizon and a decrease of 0.4 % for the 48-h horizon. The stability of this behavior is attributed to the lower emissions associated with electricity generation, primarily derived from hydro, nuclear, and wind sources in Sweden, compared to the higher emission intensity of the local district heating network, which relies on a local energy mix. The MPC generally redirects demand toward electricity use when it is economically viable, resulting in minimal increases in emissions.

The market-responsive operational cost, representing the MPC objective function, decreases by 12.7 % and 13.4 % for the 24- and 48-h horizons, respectively, indicating the effectiveness of MPC in minimizing immediate operating expenses. The total operational cost, including maintenance, total heating and electricity costs, and total system impact penalty costs, decreases by 5.1 % for the 24-h horizon and 5.4 % for the 48-h horizon, despite a marginal rise in investment costs of 1.8 % in both scenarios, likely due to the cost of the hot water tank and software for MPC implementation.

Table 4 presents the environmental and long-term financial indicators for both rule-based and MPC-controlled systems. The LCOE shows an increase of 6 % and 5 % for the 24-h and 48-h MPC cases, respectively. This unexpected trend can be attributed to a decrease in the total product energy, which is the denominator in the LCOE formula and

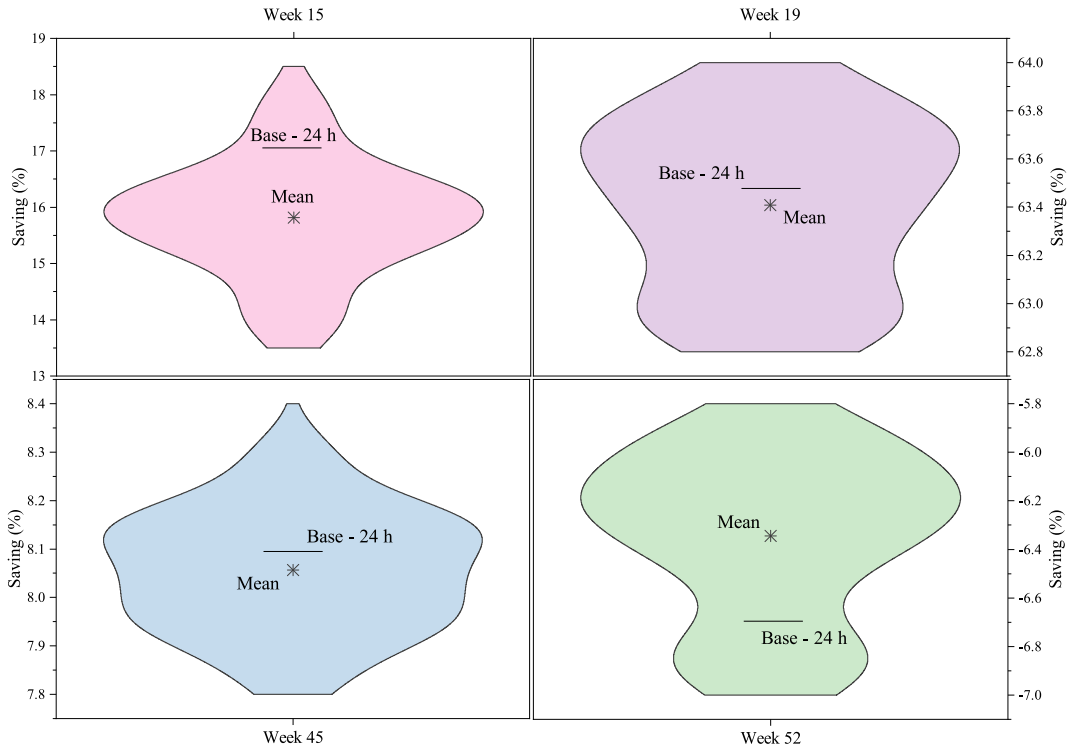


Fig. 11. Distribution of weekly operational cost savings and mean values under the 25 stochastic forecast-error scenarios (Selected weeks, horizon = 24 h).

Table 2
Cost savings (%) statistical summary under different forecast errors by week.

Week	Min	Max	Mean	Std Dev
Week 15	13.85	17.98	15.91	0.98
Week 19	62.87	63.94	63.41	0.34
Week 45	7.82	8.30	8.06	0.12
Week 52	−6.90	−5.91	−6.41	0.34

includes all useful heating and cooling outputs. MPC reduces costs by strategically minimizing the operational hours of the heat pump, particularly during periods of elevated electricity prices; this leads to a decrease in the thermal energy transferred from the heat pump to the storage tank. This reduction should be offset by more heat transferred from the district heating network; Since the increment of heat from the district heating is proportionally smaller than the decrease in energy supplied by the heat pump, an overall decrease in delivered thermal energy is expected. As a result, even with increased operational efficiency, the LCOE increases since the constant total system costs are divided by a lower energy output. The observed increase in the CO_2 index, 7.7 % for the 24-h MPC and 5.6 % for the 48-h horizon, can be attributed to the same denominator effect noted in the LCOE trend. The CO_2 index is the ratio of total CO_2 used to the net thermal energy delivered. The implementation of MPC may result in a marginal increase or decrease in CO2 usage; however, the overall useful thermal output diminishes due to the intentional reduction of the heat pump's capacity. Consequently, a greater CO_2 intensity per unit of useful energy delivered is attained.

By switching from a traditional rule-based control to MPC, the system pays for itself sooner. The payback period decreases from about 21 years to just over 17 years when MPC plans a day ahead, and to roughly 16.9 years with a two-day horizon, assuming an annual interest rate of 10 %. The difference comes from lower operational costs: MPC starts the heat pump only when it's most cost-effective and draws from the district-heating network at the optimal times. These smarter moves reduce annual operating expenses, allowing the higher upfront investment to be recovered more quickly and the system to remain more profitable over the long term. However, these results should be interpreted as indicative rather than absolute, since a sensitivity analysis of capital and tariff assumptions was not conducted. Future studies should assess how variations in electricity and heating prices influence long-term economic feasibility.

Fig. 12 illustrates that MPC significantly reduces electricity costs (spot + transmission) compared to rule-based control, particularly during transitional months. In May, costs decreased from 6188.11 SEK (rule-based) to 3519.59 SEK under the 24-h MPC, resulting in a monthly savings of 43.1 %. The 48-h MPC achieved a savings of 34.8 %. April and October exhibit significant reductions, 16.2 % and 28.0 %, respectively, with 24-h MPC, in contrast to 11.2 % and 27.5 % with 48-h MPC. The transitional months of spring and autumn present moderate heating demands, allowing MPC to enhance system operations via efficient load shifting, price-responsive control, and adaptable storage utilization. The decreased and more fluctuating demand presents enhanced opportunities for MPC to synchronize heat use with advantageous electricity pricing, all while ensuring comfort and reducing penalties.

The winter months of December and January exhibit consistent, albeit modest, cost reductions attributed to elevated and inflexible heating demand. In December, the 24-h MPC yields a 15.0 % savings, whereas the 48-h horizon achieves a 9.7 % savings. This trend highlights a notable difference: the 24-h horizon consistently outperforms the 48-h method across all active months. The reduced forecast horizon enhances the precision of electricity price and temperature predictions, facilitating improved synchronization of system operations with real-time market signals. The 48-h MPC, although effective, is somewhat limited by increased forecast uncertainty. In the summer (June–August), electricity expenses reach zero across all strategies as passive ground cooling eliminates the need for the heat pump. The findings highlight the enhanced responsiveness and cost efficiency of 24-h MPC, especially during transitional months with variable demand, where predictive control effectively utilizes market pricing dynamics and system flexibility.

Fig. 13 illustrates that heating (energy) costs consistently rise under

Table 3
Comparative techno-economic performance of the analyzed control systems under rule-based and MPC control strategies (24 h and 48 h horizons) over a year. All costs in SEK.

Heat price					
Control scenario	**Energy**	**Power**		**Total**	
Rule-based	240,445	285,000		525,445	
MPC – 24 h	252,103	285,000		537,103	
Relative increase/decrease (%)	+4.8	0		+2.2	
MPC – 48 h	247,526	285,000		532,526	
Relative increase/decrease (%)	+2.9	0		+1.3	
Electricity price					
Control scenario	**spot**	**fixed**	**transmission**	**power**	**Total**
Rule-based	74,865	7800	15,815	39,360	137,840
MPC – 24 h	66,140	7800	13,613	39,360	126,913
Relative increase/decrease (%)	−11.7	0	−14	0	−8
MPC – 48 h	68,147	7800	14,039	39,360	129,346
Relative increase/decrease (%)	−9	0	−11.2	0	−6.2
System impact penalty					
Control scenario	**CO_2 emission**	**Peak shaving**	**TES preservation**	**Comfort**	**Total**
Rule-based	32,749	30,959	39,182	17,597	120,487
MPC – 24 h	32,974	23,967	5291	18	62,250
Relative increase/decrease (%)	+0.7	−22.6	−86.5	≈ −100	−48.3
MPC – 48 h	32,627	25,182	3762	24	61,595
Relative increase/decrease (%)	−0.4	−18.7	−90.4	≈ −100	−48.9
Total price					
Control scenario	**Investment**	**Maintenance**	**Market-responsive operational**	**Total operational**	**Total**
Rule-based	5,013,518	250,676	451,612	1,034,448	6,047,966
MPC – 24 h	5,103,518	255,176	394,106	981,442	6,084,960
Relative increase/decrease (%)	+1.8	+2	−12,7	−5.1	≈ 0
MPC – 48 h	5,103,518	255,176	391,307	978,643	6,082,161
Relative increase/decrease (%)	+1.8	+2	−13,4	−5.4	≈ 0

Table 4
Comparative environmental and long-term indicators of the analyzed control systems under rule-based and MPC control strategies (24 h and 48 h horizons) over a year.

Control scenario	LCOE (SEK. MWh^{-1})	CO_2 used (Tonne)	CO_2 index ($kg.MWh^{-1}$)	Payback (Year)
Rule-based	5142	27.3	23.2	21.4
MPC – 24 h	5453	27.5	25	17.1
Relative increase/decrease (%)	+6	+0.7	+7.7	−20
MPC – 48 h	5396	27.2	24.5	16.9
Relative increase/decrease (%)	+5	−0.4	+5.6	−21

MPC compared to rule-based systems. In January, heating costs increase from 36,478 SEK to 36,815 SEK for 24 h and 36,845 SEK for 48 h. In December, costs rose from 38,473 SEK to 44,056 SEK and 41,964 SEK, respectively. The increases are particularly pronounced during shoulder months such as April and May, where MPC results in significant cost escalations, up to 39 % in April and 85 % in May. MPC aims to minimize market-responsive operational costs, including heating (energy) prices, electricity costs (including spot and transmission), and various system impact penalties, specifically CO_2 emissions, peak shaving, thermal energy storage preservation, and comfort deviations. Heating energy costs may rise (as shown in Fig. 13) due to the load shifts from heat pumps to the district heating network; however, such decisions frequently result in significant reductions in other cost components. Fig. 12 illustrates a significant reduction in electricity costs under MPC, particularly during periods of price volatility, resulting from optimal load shifting and decreased exposure to peak rates. In this context, an increase in the heating term may be offset by a more significant reduction in the electricity and penalty terms, resulting in a lower total market-responsive operational cost.

September, however, is distinguished by a cost reduction under MPC: the heat pump is switched off, a dedicated ground-recovery mode is adopted, heat is extracted from the district heating network and injected into the boreholes to restore thermal balance, and a strong positive absolute saving is reported in Fig. 13(b), contrasting with the negative savings recorded in other months. This highlights the effectiveness of MPC in managing thermal energy storage preservation for the remainder of the year. Through the strategic optimization of borehole utilization in previous seasons, MPC effectively reduces the heat from the network in September, thereby achieving the desired average ground temperature and lowering costs during the recharge phase.

The figure finally depicts that the minimal difference observed between 24-h and 48-h MPC results, both in seasonal and annual trends, underscores the robustness and reliability of the control strategy. The system exhibits comparable performance across two distinct forecasting horizons, suggesting that MPC decisions are consistent despite longer-term uncertainties. This consistency reinforces the feasibility of implementing MPC in practical scenarios, where accurate predictions may not always be assured.

Fig. 14 presents the monthly system impact penalties associated with rule-based control and MPC utilizing various prediction horizons. Penalties cover costs related to CO_2 emissions, peak shaving, borehole thermal energy storage preservation, and comfort violations. A consistent reduction in penalties is evident across nearly all months with the application of MPC. In May, a notable instance occurred where penalties decreased significantly from 3539 SEK under rule-based control to 385 SEK (48 h) and 384 SEK (24 h), representing a reduction of nearly 90 %. Significant reductions in penalties are observed in October, decreasing from 1278 SEK to 354 SEK and 363 SEK, as well as in April, where the MPC lowers penalties from 2027 SEK to roughly 647–696 SEK. These underscores MPC's capability during transitional months, where system flexibility facilitates accurate load shifting and TES optimization, thereby preventing peak operations and enhancing comfort acceptance.

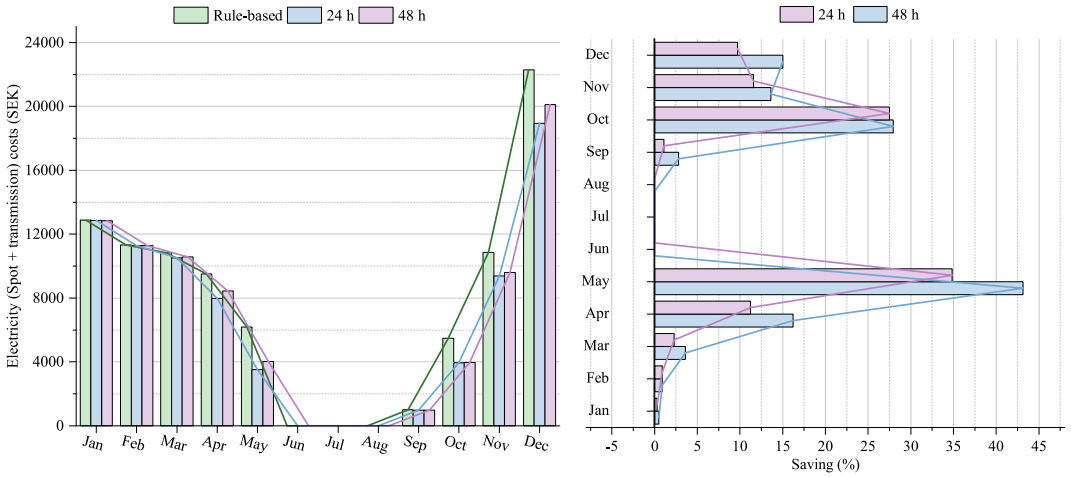


Fig. 12. Monthly electricity costs (spot + transmission) and savings under rule-based and MPC control strategies (24 h and 48 h horizons). Units in SEK. Comparison based on the same operational period to highlight the dynamic tariff response under each control horizon.

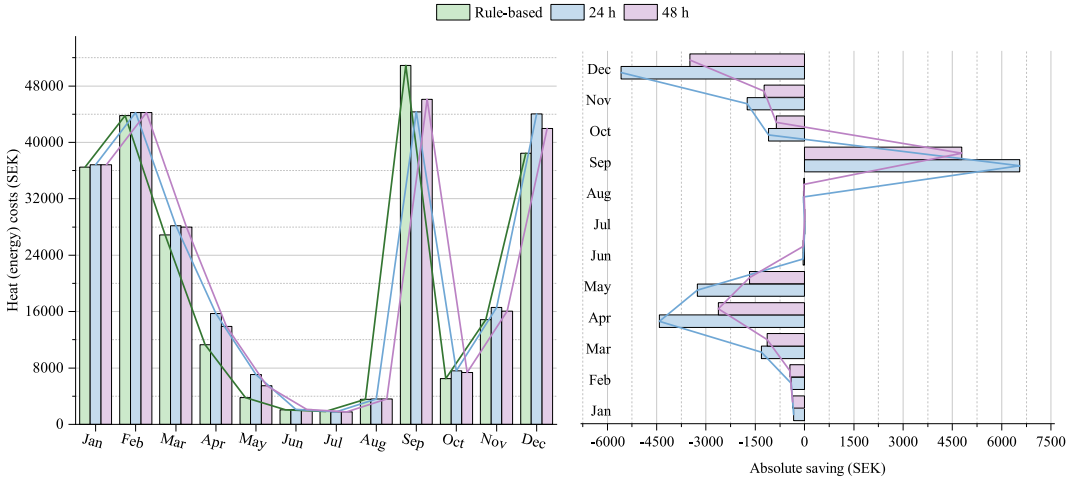


Fig. 13. Monthly heating costs (energy) and savings under rule-based and MPC control strategies (24 h and 48 h horizons).

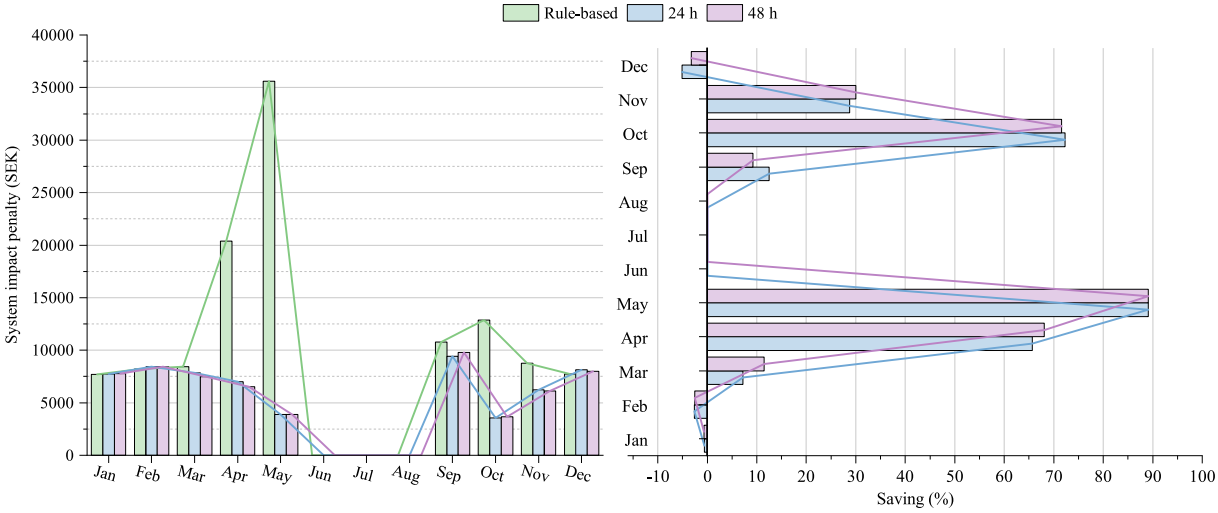


Fig. 14. Monthly system impact penalty and savings under rule-based and MPC control strategies (24 h and 48 h horizons).

During the winter months of January, February, and December, penalties exhibit relative stability across control strategies, reflecting limited flexibility attributed to increased demand and system saturation. The minor variations between 24-h and 48-h outcomes underscore the reliability and robustness of the control strategy across different time frames.

Fig. 15 demonstrates the monthly market-responsive operational cost of rule-based and MPC control strategies. This metric combines heating (energy) costs, electricity spot and transmission prices, and system impact penalties, functioning as the primary objective function minimized by MPC. The figure illustrates that, despite rising individual components, such as heating energy costs (Fig. 13), MPC consistently achieves a significant reduction in total market-responsive operational costs, especially during the shoulder season.

Significant cost reductions are observed in May, with market-responsive operational costs decreasing from 45,607 SEK (rule-based) to 14,463 SEK (24 h) and 13,405 SEK (48 h), indicating savings exceeding 70 %. Similarly, in April, costs decreased from 41,182 SEK to 30,672 SEK (24 h) and 28,866 SEK (48 h). The reductions result mainly from optimized electricity utilization and reduced system penalties, particularly for TES preservation and peak shaving, facilitated by MPC's proactive and price-responsive control. In October and November, savings are significant, with market-responsive operational costs reduced by approximately 40 % under MPC. Conversely, January, February, and December exhibit negligible variations between rule-based and MPC costs. This outcome indicates limited operational flexibility due to consistently increased heating demands and reduced opportunities for load shifting or penalty mitigation. The close alignment of 24-h and 48-h results across all months demonstrates the high reliability and consistency of the MPC strategy. Despite varying forecast durations, the small performance disparity between the two horizons indicates that the controller continues to function effectively in the face of greater uncertainty regarding future conditions.

Fig. 16 illustrates the progression of the net present value over 30 years for the system utilizing rule-based control and MPC strategies, taking into account investment costs, operational costs, and financial returns. At year 0, all three strategies commence with an initial investment of approximately 5.1 million SEK, with slight differences due to the addition of hot water tank and controller expenses. The accumulation of operational savings over time leads to an increase in NPV, with the point at which it becomes positive indicating the payback period.

The rule-based control scenario achieves break-even at year 21, whereas the MPC strategies demonstrate a significantly earlier payback, at year 17 for the 24-h horizon and year 16.9 for the 48-h horizon. This early payback illustrates the long-term economic benefits of predictive control, resulting from decreased electricity costs and reduced system penalties, despite elevated heating (energy) expenses in certain months.

The figure further demonstrates that by year 30, the profitability gap expands significantly. The rule-based control results in a final NPV of 416,725 SEK, while the 24-h MPC and 48-h MPC yield significantly higher returns of 868,801 SEK and 895,260 SEK (with nearly double the net return), respectively. The findings indicate that MPC enhances operational efficiency and provides better long-term financial performance, thereby validating the slightly increased initial investment necessary for MPC setup, such as control software and thermal storage sizing. The similar NPV curves of the 24-h and 48-h horizons demonstrate the effectiveness of MPC strategies; the system maintains economic resilience and consistently generates substantial financial returns throughout its lifecycle, even with longer forecast periods.

Fig. 17 presents a comparison of the monthly CO_2 emission index (kg/MWh) for the system utilizing rule-based control and model predictive control with 24-h and 48-h horizons, alongside the emission intensities of the electricity mix and district heating data by local providers in the city of Uppsala. This index quantifies emissions per unit of useful energy delivered, thereby reflecting the environmental performance of the control strategies.

Throughout all months, both MPC strategies consistently demonstrate lower CO_2 intensity than the rule-based control, underscoring

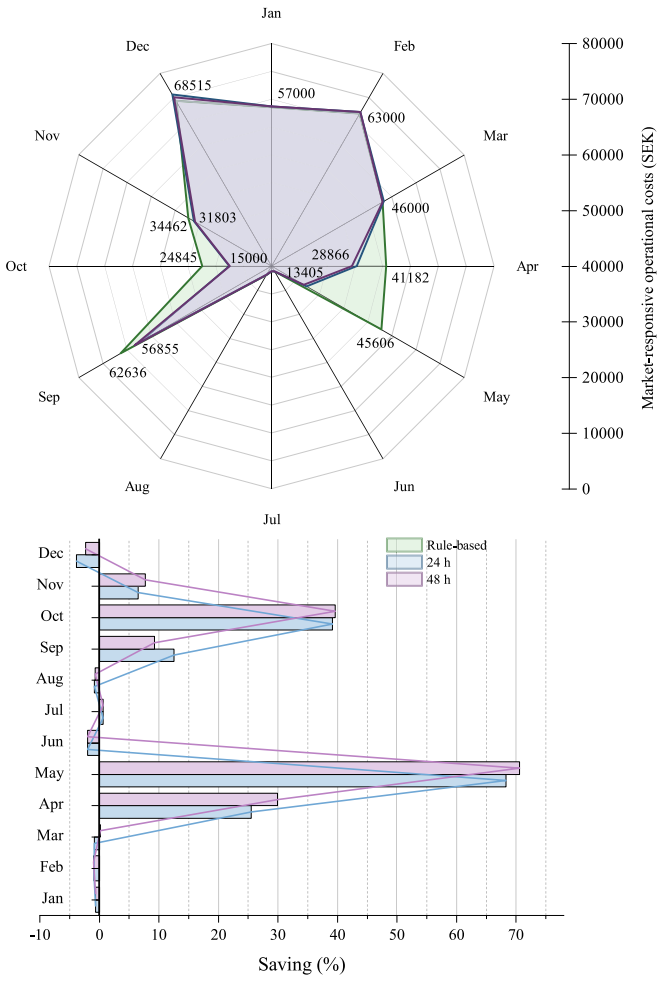


Fig. 15. Monthly market-responsive operational costs and savings under rule-based and MPC control strategies (24 h and 48 h horizons).

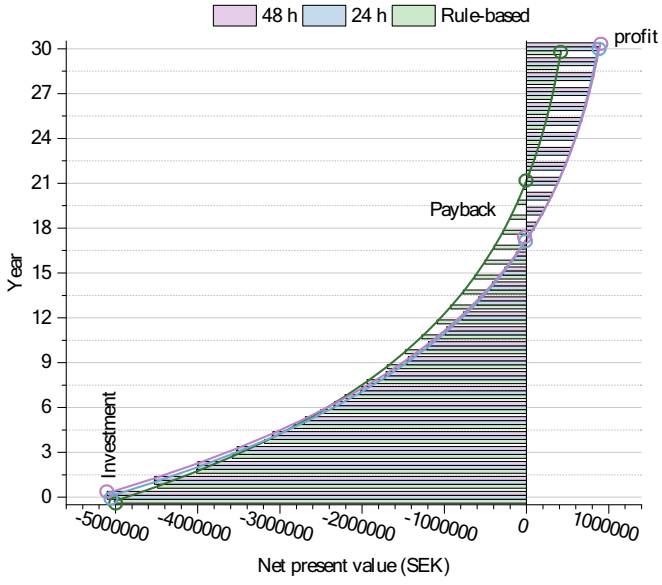


Fig. 16. Variation of net present value over 30 years under rule-based and MPC control strategies (24 h and 48 h horizons).

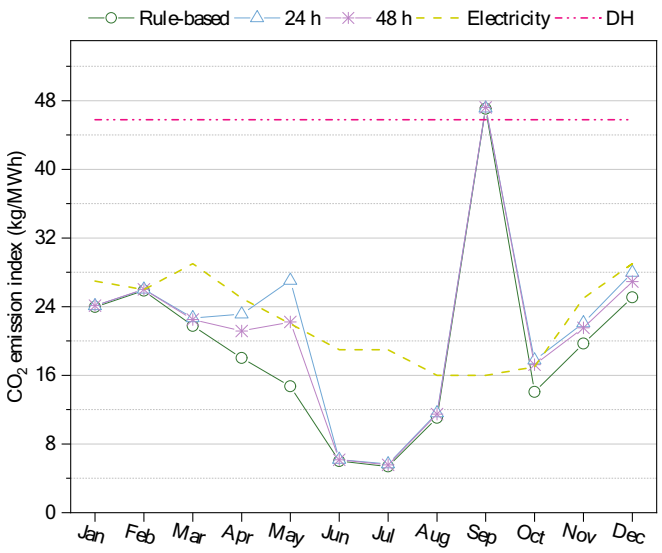


Fig. 17. Comparison of monthly CO_2 emission index under rule-based and MPC controllers alongside emission intensities from local heating and electricity providers [43,45].

their effectiveness in reducing carbon impact. In May, the emission index decreases from 14.73 kg/MWh (rule-based) to 12.21 kg/MWh (48 h MPC) and 12.04 kg/MWh (24 h MPC). Comparable reductions occur in March, April, October, and November, during which MPC realizes 1–2 kg/MWh emissions lower than the rule-based framework.

In September, the CO_2 index reaches its maximum for all strategies, ranging from 47 to 47.2 kg/MWh, surpassing the DH baseline. This demonstrates the deliberate application of the district heating network to introduce heat into the boreholes, ensuring long-term thermal stability. This process is essential yet intensive in terms of emissions. Nonetheless, MPC demonstrates superior performance to the rule-based approach by approximately 1 kg/MWh, indicating enhanced scheduling and emission-aware operation even under constrained conditions.

An important observation is that the entire system consistently operates below the CO_2 intensity of the district heating network (45.8 kg/MWh) throughout the year. This underscores the benefit of the hybrid configuration, which utilizes the borehole TES and a low-carbon electricity mix in Uppsala, primarily derived from hydropower and wind, to operate the heat pump during times when it is more environmentally advantageous than extracting heat from the combined heat and power-based district heating network.

However, analyzing these results within the relevant energy system framework is crucial. Sweden's electricity mix ranks among the cleanest worldwide, exhibiting average emissions of below 30 gCO_2/kWh in typical years; however, seasonal and peak-time fluctuations result in increased reliance on higher-intensity imports. During winter peak demand or regional shortages, electricity may be imported from adjacent Nordic or European grids, which have carbon intensities of approximately 70–100 g CO_2/kWh and 300 g CO_2/kWh, respectively. District heating systems, by contrast, utilize locally governed fuel mixes and generally demonstrate more stable carbon characteristics over time. Therefore, this study demonstrates that MPC effectively utilizes the clean Swedish electricity mix to lower emissions; however, such advantages depend on the cleanliness of both domestic and imported electricity sources. It is essential to avoid generalizing the environmental advantages of electricity-based heating solutions without considering system-level dynamics, temporal variations, and regional grid dependencies.

6. Conclusion, outlook, and future work

This study investigates whether model predictive control can deliver further performance improvements when applied to an already ultra-energy-efficient real case building thermal system integrating boreholes, ground-source heat pumps, and smart ventilation. According to Monte Carlo analysis, the proposed MPC framework demonstrates strong robustness under realistic forecast uncertainties in outdoor temperature and electricity prices. The results confirm that MPC consistently delivers improvements across multiple dimensions even in systems that are already optimized through advanced rule-based control logic. From an operational perspective, MPC achieves reductions of up to 13.4 % in market-responsive operational costs, an increment of the net present value by about 20 % over a 20-year horizon, and a shortening of the payback time from 21.4 years to around 17 years compared to existing well-established rule-based control. MPC also improves the thermal comfort while reducing the borehole TES temperature degradation and peak power penalties by more than 90 % and 20 %, demonstrating the importance of designing a multi-layer cost function that considers not only immediate costs but also long-term facets, thermal balance sustainability, and demand-side flexibility under dynamic market and environmental conditions. However, modest increases in heating costs (+2 %) and a normalized LCOE (+6 %) are noted as a result of optimized scheduling and a reduction in overall energy delivered, highlighting the strategic switching ability of MPC between electricity and heat sources. This facilitates adaptation to tariff structures and reduces overall costs.

According to the seasonal variations, system flexibility plays a key role in the performance of MPC. During the transitional months (March–May), when loads change more frequently and operating modes are more diverse, MPC has the highest savings by utilizing load shifting and storage capacity. In contrast, during winter, MPC saves less because heating demand increases, resulting in a lower degree of system freedom. This context-dependent performance shows that MPC effectiveness is shaped not only by algorithmic design but also by the physical flexibility and thermal inertia of the proposed system. Environmentally, applying MPC, the proposed thermal system has a lower CO_2 emission index throughout the year than both the existing rule-based controller and the district heating baseline. The annual CO_2 emissions decrease by around 30 %, and the CO_2 index increases by 35 %, thanks to strategic employment of various energy sectors and Sweden's low-carbon electricity mix. However, these environmental benefits remain country-specific. In nations characterized by a carbon-intensive electricity grid, reductions in emissions may be reduced or potentially reversed during peak import periods. Therefore, the environmental advantage of MPC is based upon the local energy mix and must be assessed within the context of regional decarbonization efforts.

It is worth noting that, although the MPC was implemented within a co-simulation framework, the building model was validated against measured data. Therefore, the results reflect a real building case study rather than a full-scale deployment. On the technical side, further improvements to the existing MPC framework could be achieved through integration with reinforcement learning, which would continuously update system feedback and refine performance over time. From an experimental perspective, field validation using advanced sensor networks and real-time data infrastructure will be essential for integrating simulation-based performance with actual operational conditions. Additionally, comprehensive research comparing MPC-based frameworks across multiple countries with diverse energy environments, each characterized by unique electricity mixes, tariff structures, and climatic conditions, would yield significant insights into their transferability and scalability.

CRediT authorship contribution statement

Amirmohammad Behzadi: Writing – original draft, Visualization,

Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Naser Goudarzi:** Software, Methodology, Investigation, Formal analysis. **Adnan Ploskic:** Writing – review & editing, Validation, Supervision, Methodology, Data curation. **Eva Thorin:** Writing – review & editing, Validation, Supervision, Resources, Data curation. **Sasan Sadrizadeh:** Writing – review & editing, Validation, Supervision, Software, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Appendix

Table A1 describes the components and their types, along with detailed descriptions in TRNSYS, to facilitate the thermodynamic modeling of the proposed smart thermal energy system.

Table A1
TRNSYS component types and their corresponding descriptions.

Component	TRNSYS type	Description
Borehole TES	577	A vertical double U-tube ground heat exchanger is represented by 25 boreholes, each 250 m deep and 6 m apart, through which a 30 / 70 ethylene-glycol / water fluid circulates to draw heat in winter and reject it in summer; heat transfer within the tubes and adjacent soil is simulated by a hybrid model in which global finite-difference calculations describe long-term ground dynamics, local solutions capture borehole interactions, and a steady-state flux covers constant loads, while grout, pipe, and fluid resistances are assigned from field and manufacturer data to ensure realism.
Heat pump	827	Non-dimensional performance maps derived from catalog data represent a two-stage, water-to-water heat pump (NIBE F1345). Heating capacity and power input are evaluated against brine-inlet temperature, supply-water set-point, and mass-flow rate, enabling year-round dynamic analysis. The evaporator is coupled to a 30 / 70 ethylene-glycol-water borehole loop; fluid temperatures may drop below 0 °C in winter, yet frost losses are avoided because no air-side exchange occurs, so defrost cycles are excluded.
Energy wheel	667	The energy wheel operates as a rotary heat exchanger through which both sensible and latent heat are recovered from the exhaust airstream to pre-condition the incoming fresh air. This component is essential for limiting ventilation-related heating and cooling loads, especially in cold-climate regions such as Sweden, where outdoor temperatures fall sharply during winter.
Heating coil (AHU)	753	A hot-water air-heating process is simulated through the bypass-fraction method. Warm water is circulated within the coil, allowing part of the incoming air to be brought into thermal equilibrium with the coil's mean fluid temperature, while the remainder is routed past the coil unchanged. Heat transferred to the airstream is then calculated from the bypass fraction and the log-mean temperature difference.
Cooling coil (AHU)	508	The cooling coil is modeled using the bypass-fraction method, in which a defined portion of the incoming air stream bypasses the coil while the remainder is cooled to the coil's mean fluid temperature and exits at saturation, representing the maximum latent and sensible cooling potential. Two operating modes are provided: an uncontrolled mode that removes as much heat and moisture as the inlet conditions allow, and a controlled mode that diverts part of the fluid or air stream to keep the supply-air temperature below a specified limit.
Hot water tank	158	Type 158 depicts a vertical, constant-volume storage tank partitioned into user-defined isothermal nodes to capture stratification. Each node exchanges heat with its neighbours through conduction and flow-induced or buoyancy-driven mixing, while surface losses from the top, bottom, and sides are included. Up to two fluid streams may enter or exit the tank, and optional auxiliary heating can be applied.
Heat exchanger	91	A steady-state, design-independent liquid-to-liquid heat exchanger is represented, exchanging heat between the borehole circuit and the internal building loop rather than an air coil. A fixed-effectiveness method is employed, so that the outlet temperatures are determined by the prescribed effectiveness and the smaller of the two streams' heat-capacity rates.
Feedback controller	22	The iterative feedback controller holds A designated system variable at its target value; an input is continually adjusted until the desired output is reached. This approach is especially advantageous in HVAC applications, where interdependent variables and nonlinear behavior render simple control logic inadequate.
PID controller	23	Dynamic responses in time-dependent components, such as pumps, heat pumps, and valves, are regulated by the proportional–integral–derivative (PID) controller. The measured variable (e.g., indoor or supply-fluid temperature) is continuously compared with its set-point, and the control signal is adjusted to minimize the error. Stable operation is thereby maintained, overshoot is limited, and energy efficiency is enhanced while thermal comfort and energy objectives are satisfied.
Fan	744	A variable-speed fan or blower operating against a specified pressure drop is simulated, with electric power consumption and airflow delivery calculated from the prescribed fan curve, which is typically expressed as a quadratic relationship between airflow and pressure rise.
Pump	110	Water is circulated through the boreholes, heat exchangers, and coils by variable-speed pumps. The model estimates the electric power needed to maintain the specified mass-flow rate against the given pressure drop for each time step.

Table A2 lists the key parameters needed to model the building, the proposed smart HVAC system, and their associated components. The circulating heat transfer fluid in the borehole system is a 30/70 ethylene glycol-water mixture, chosen for its antifreeze characteristics and suitability for sub-zero ground temperatures. The building's façade composition features a 30 % window-to-wall ratio on the north side and 26 % on the east side, while the southwest façade exhibits 28 % glazing; the remaining sections are composed of concrete. Occupancy-related internal gains were modeled separately for each zone, with occupant density set at 6 m^2/person for the restaurant and 17 m^2/person for the offices. Heat gains from occupants were 19.7 W/m^2 and 7 W/m^2, respectively. Equipment loads added 4 W/m^2 in the restaurant and 12 W/m^2 in the offices, whereas lighting loads were 20 W/

m^2 in the restaurant and 12 W/m^2 in the office zones.

Table A2
Thermo-physical and heat transfer parameters of building and thermal energy system components.

Variable	Unit	Value	Variable	Unit	Value
Specific heat capacity of the mixture	J/(kg·K)	3600	Heating capacity of heat pump (0/35 °C)	kW	39.94
Density of the mixture	kg/m^3	1048	Supplied power of the heat pump (0/35 °C)	kW	8.9
Borehole depth	m	250	COP (0/35 °C)	–	4.49
Borehole numbers	–	25	Heating capacity of heat pump (0/45 °C)	kW	38.9
Borehole radius	mm	70	Supplied power of heat pump (0/45 °C)	kW	10.61
Borehole space	m	6	COP (0/45 °C)	–	3.67
Pipe radius	mm	16/14	Heating capacity of heat pump (10/35 °C)	kW	51.71
Fill thermal conductivity	W/(m·K)	0.8	Supplied power of the heat pump (10/35 °C)	kW	9.81
Pipe thermal conductivity	W/(m·K)	0.38	COP (10/35 °C)	–	5.27
Zone's set point (summer)	°C	24	Heating capacity of heat pump (10/45 °C)	kW	50.79
Zone's set point (winter)	°C	22	Supplied power of the heat pump (10/45 °C)	kW	11.82
Air change (restaurant)	1/h	6	COP (10/45 °C)	–	4.3
Air change (offices)	1/h	2	Brine maximum pressure	MPa	0.6
Heat coefficient of windows	W/(m^2·K)	1	Minimum flow rate of brine	l/s	1.59
Heat coefficient of roofs	W/(m^2·K)	0.65	Nominal flow rate of brine	l/s	2.09
Heat coefficient of floors	W/(m^2·K)	0.65	Minimum flow of heating medium	l/s	0.64
Heat coefficient of walls	W/(m^2·K)	0.6	Nominal flow of heating medium	l/s	0.93

Fig. A1 illustrates the hourly volumetric flow rate schedule for the three ventilation lines that deliver conditioned air to the building zones. Lines 1 and 2 are connected to the office areas, whereas Line 3 is designated for the restaurant. Office ventilation systems follow standard operational schedules on weekdays, with Line 1 functioning continuously at a base flow of 5 m^3/s, increasing to 7.5 m^3/s during occupied hours from 6:00 to 20:00. Line 2 is inactive on weekends, indicating the unoccupied status of the associated office zones, and operates only on weekdays, following the same hourly flow pattern as Line 1. Line 3, focused on the restaurant, features a dynamic and diverse schedule that adapts to fluctuations in occupancy and operational hours during the week. The flow rate for Line 3 reaches a maximum of 4.2 m^3/s on most weekdays, exhibiting significant fluctuations on Fridays and weekends, characterized by decreased evening operations and total shutdown during early morning hours. This scheduling strategy accurately represents the occupant-driven air handling demand of the proposed smart system and promotes energy efficiency by reducing primary energy via load-responsive operation. Regarding the working hours, office areas are operated from 07:00 to 18:00 on weekdays and kept closed on weekends, whereas the restaurant is run from 08:00 to 18:00 on Monday and Tuesday, 08:00 to 20:00 on Wednesday through Friday, and 11:00 to 22:00 on Saturday, remaining closed on Sunday; as a result, zone-specific ventilation, heating, and cooling loads are set according to these occupancy schedules.

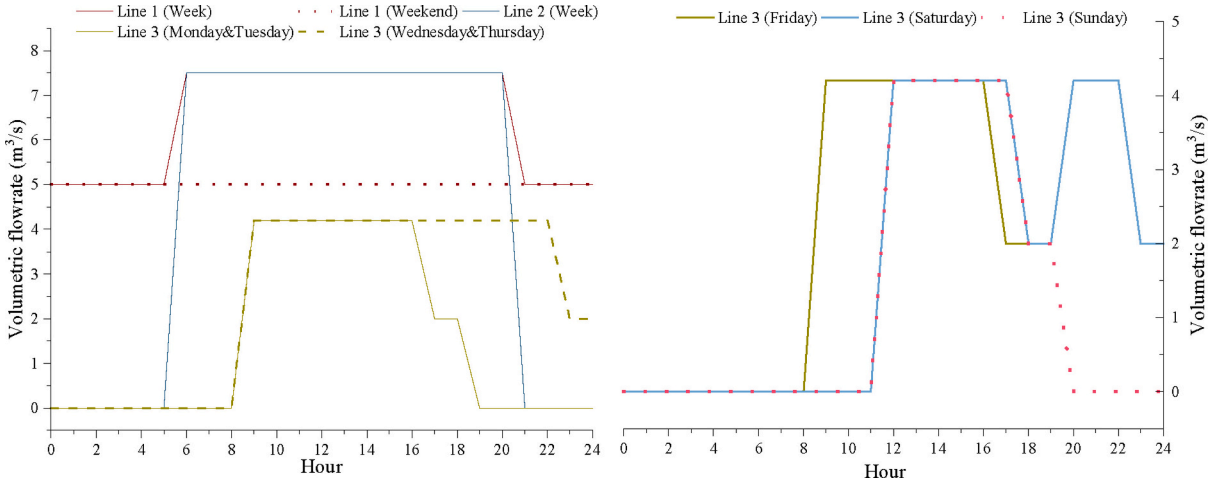


Fig. A1. The volumetric flow rate schedule of air entering each ventilation line.

Fig. A2(a) presents the statistical distribution of the introduced forecast errors in outdoor temperature, represented as zero-mean Gaussian noise with a standard deviation of 1 °C, over a 24-h time horizon. The histogram, enhanced by a kernel density estimate, validates the anticipated bell-shaped, symmetric distribution centered at 0 °C. The shaded regions represent intervals of standard deviation. Approximately 68 % of the errors are within ±1 °C, about 95 % are within ±2 °C, and nearly 99.7 % are within ±3 °C. The rug plot under the histogram visually represents the distribution and density of individual simulation data points. This distribution reflects real-world uncertainty patterns in short-term weather forecasts and serves as a foundation for assessing the sensitivity of the MPC to prediction inaccuracies. The mean error being zero indicates that the errors are unbiased, meaning the forecast has an equal probability of overestimating or underestimating actual conditions, which ensures a rigorous, robust test for the MPC.

Fig. A2(b) illustrates the distribution of forecast errors in hourly spot electricity prices, modeled as a normal distribution with a mean of zero and a standard deviation of 0.03 SEK/kWh over a 24-h time horizon. The histogram displays a symmetric bell-shaped curve that aligns with Gaussian features, indicating that the forecast errors were unbiased and randomly distributed. The shaded areas denote confidence intervals: 68 % of values are within ±0.03 SEK/kWh, 95 % within ±0.06 SEK/kWh, and 99.7 % within ±0.09 SEK/kWh. The kernel density estimate overlay reinforces the

observed symmetry, and the rug plot below offers insights into the density and distribution of individual data points across all 25 scenarios. This figure accurately depicts short-term forecasting noise in actual electricity markets, particularly relevant to systems in Sweden, which are characterized by volatile spot pricing. The controlled stochastic variation facilitates a rigorous assessment of MPC's response to economic optimization under imperfect price predictions, which is particularly significant for load shifting and peak shaving strategies.

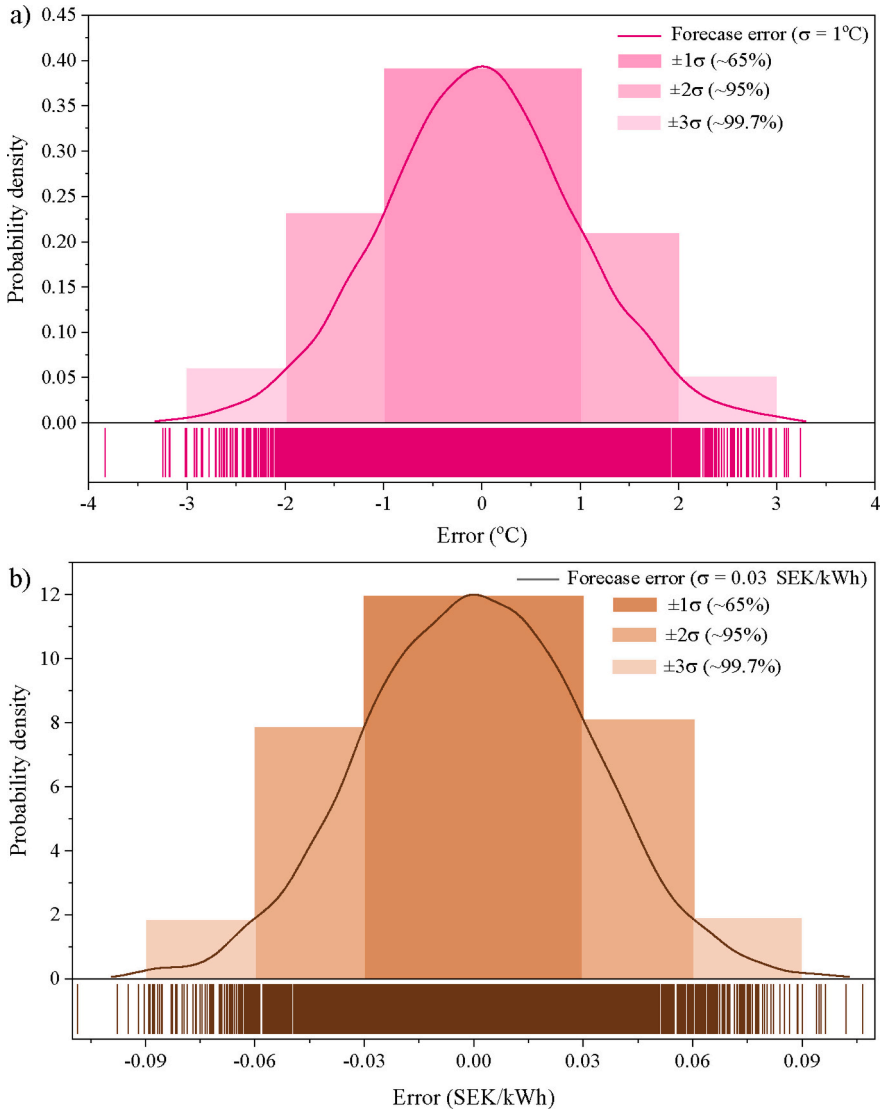


Fig. A2. The histogram of a) outdoor temperature (($\epsilon_T \sim N(0, 1\ °C)$)) and b) electricity price (($\epsilon_P \sim N(0, 0.03\ SEK/kWh)$)) forecast errors across 25 scenarios (Week 15, Horizon = 24 h).

Data availability

Data will be made available on request.

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